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The Evolution of Japan's Space Security Policy and a Blueprint for Strengthening the U.S.-Japan Space Security Partnership

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Kari A. Bingen

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A Report of the CSIS Aerospace Security Project

CSIS | CENTER FOR STRATEGIC &
INTERNATIONAL STUDIES

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The analysis and conclusions contained in this report are solely those of the authors and any errors or omissions remain the exclusive responsibility of the authors.

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Acronyms and Abbreviations

ASAT	Anti-satellite
BMD	Ballistic missile defense
C2	Command and control
CIRO	Cabinet Intelligence and Research Office
CSICE	Cabinet Satellite Intelligence Center
DOD	Department of Defense
EO	Earth observation
FY	Fiscal year
GEO	Geostationary orbit
GPS	Global Positioning System
GSSAP	Geosynchronous Space Situational Awareness Program
HGV	Hypersonic glide vehicle
IC	Intelligence community
IGS	Intelligence-Gathering Satellite
INDOPACOM	Indo-Pacific Command
IR	Infrared
ISAS	Institute of Space and Aeronautical Science
ISRT	Intelligence, surveillance, reconnaissance, and tracking
ISS	International Space Station
ITAR	International Traffic in Arms Regulations
J6	Joint Staff C4 Systems Department
JASDF	Japan Air Self-Defense Force
JAXA	Japan Aerospace Exploration Agency

JCO	Joint Commercial Operations
JSOG	Japan Space Operations Group
LEO	Low Earth orbit
MAFF	Ministry of Agricultural, Forestry, and Fisheries
MELCO	Mitsubishi Electric Corporation
MEO	Medium Earth orbit
METI	Ministry of Economics, Trade Industry
MEXT	Ministry of Education, Culture, Sports, Science, and Technology
MHI	Mitsubishi Heavy Industries
MIC	Ministry of Internal Affairs and Communications
MLIT	Ministry of Land, Infrastructure, Transport, and Tourism
MOD	Ministry of Defense
MOE	Ministry of the Environment
MOFA	Ministry of Foreign Affairs
MTCR	Missile Technology Control Regime
MW/MT	Missile warning and missile tracking
NAL	National Aerospace Laboratory of Japan
NASDA	National Space Development Agency
NDS	National Defense Strategy
NSpC	National Space Council
NSPS	National Space Policy Secretariat
NSS	National Security Strategy
OPIR	Overhead persistent infrared
PWSA	Proliferated Warfighter Space Architecture
QZSS	Quasi-Zenith Satellite System
R&D	Research and development

SAA	Space Activities Act
SAR	Synthetic aperture radar
SATCOM	Satellite communications
SDA	Space domain awareness
SDF	Self-Defense Force
SLV	Space launch vehicle
SSA	Space situational awareness
SSC	Space Systems Command
SSI	Space Security Initiative
TSA	Technology Safeguards Agreement
USSF	United States Space Force
USSPACECOM	U.S. Space Command

Executive Summary

The last few years have seen a transformational change in Japan's approach to space security, with space capabilities now featured prominently in Tokyo's national security and defense strategies. While Japan has long been considered one of the world's most advanced spacefaring nations, its accomplishments have occurred largely in civil space exploration and science, with Japanese government policy ascribing nonmilitary purposes for space activities. However, intensifying threats in the region have led to a new emphasis on leveraging the space domain for national defense and the need for deeper partnerships at the policy, operational, and industrial levels to deliver space capabilities on timelines paced to the threat.

While Japan's 2008 Basic Space Law created the legal basis for the country to pursue national security space activities, its 2022 security documents provided the impetus for "radically expand[ing] the use of space systems for national security."¹ Sitting in the middle of a complex Indo-Pacific security environment, Japan's regional security concerns are dominated by intensifying Chinese and North Korean military activities, while Russia's invasion of Ukraine presages what aggression in the region could look like. Tokyo has set 2027 as a key milestone for strengthening its national defense, with space capabilities as one part of a broader strategic effort that includes force modernization, munitions acquisitions, infrastructure improvements, and enhanced training and exercises.²

Japan's 2022 National Security Strategy (NSS), National Defense Strategy (NDS), and Defense Buildup Program, together with its 2023 Space Security Initiative (SSI), designate key priorities for future space acquisitions, operational activities, and partnerships necessary to advance its

national and economic security interests. The SSI outlines the country's national security space architecture plans. This ambitious vision includes satellites that can detect and track missiles, including hypersonic glide vehicles (HGVs); intelligence, surveillance, reconnaissance, and tracking (ISRT) satellites to provide insights on land and maritime threats and targeting information for counterstrike systems; space-based communications satellites; positioning and navigation satellites; and space domain awareness systems, all supported with a suite of government and commercial space launch vehicles.³

Funding to expand Japan's space capabilities and grow its industrial base is following the plans. Tokyo has more than doubled its investments across civil and national security space over the last five years, including in defense-specific and dual-use space technologies, and plans further increases. Over the next 10 years, the Japanese government has committed ¥1 trillion (\$6.9 billion) toward a Space Strategy Fund aimed at expanding Japan's space industrial base that the Japan Aerospace Exploration Agency (JAXA) will administer.⁴ Also, over the next five years, an additional ¥1 trillion is anticipated to be allocated for space security.⁵ Like the United States and other countries, Japan recognizes the economic and technological benefits of space and how government investments create a demand signal to spur private sector innovation and attract technical talent. It also sees advances in the space domain as helping to solve global sustainability, resource management, and disaster relief challenges here on Earth.⁶

Organizational reform is also following the plans. The Japan Air Self-Defense Force (JASDF) will be renamed the Japan Air and Space Self Defense Force in 2027 to “reinforc[e] its space operation capability and . . . [develop] a system to ensure superiority in use of space.”⁷ Within the JASDF, the Japan Space Operations Group (JSOG) was created in 2020 as the first space-focused unit that will continue to grow as it takes on more operational space missions.



President Joe Biden and Prime Minister Kishida Fumio shake hands during the Japan-U.S. summit meeting on May 23, 2022, in Tokyo, Japan.

Photo: David Mareuil/Getty Images

Japan's approach to security remains strongly tethered to the policies and actions of the United States, with deepening bilateral ties “at a concrete level” being “of vital importance.”⁸ Cooperation and collaboration, particularly with the United States, is an integral part of Japanese strategy in the space domain. Likewise, expanding cooperation with allies and partners has been a pillar of U.S. national space policy across successive administrations.⁹ In May 2024, the assistant secretary of defense for space policy, Dr. John F. Plumb, characterized such partnership as “an asymmetric advantage that our competitors can never hope to match” that advances collective security,

underpins military strength, and contributes to the safety and sustainability of the space domain.¹⁰

Space has become more prominent in high-level discussions between the two nations, including at the president and prime minister level, cabinet and ministry level, and senior military leader level. U.S. and Japanese space leaders convene regularly through a comprehensive space dialogue, the two countries exchange military liaison officers, and business leaders from both countries now participate in dialogues to discuss U.S.-Japan space industry collaboration.¹¹

Key Observations

A few broad observations stood out to the authors throughout the course of their research:

- **Alignment of interests:** With a shared understanding of the threat environment, alignment of strategic interests, strong leadership and political support, and commitment of personnel and resources, there is now a window to advance a comprehensive and enduring agenda for bilateral cooperation on space security. While the authors focused heavily on security cooperation, cooperation across other instruments of national power, such as economic and diplomatic activities, were also recognized as essential to advancing both countries' national security interests.
- **National security space proficiency and capacity:** Japan grew into one of the world's most advanced spacefaring nations through its civil space programs; however, past government policy mandating that the government of Japan utilize space only for nonmilitary purposes led to its defense institutions and related industrial base having less exposure to, and experience with, space missions. Further, it resulted in a limited market for Japan's space industry to serve. While the current emphasis on space security creates greater opportunities for U.S.-Japan space cooperation, it also—understandably so—highlights areas where Japan's defense institutions and industry need to build greater space proficiency, operational experience, and industrial capacity, especially in building and operating large satellite constellations for security missions.
- **Ground architectures and satellite constellations:** Many of the authors' discussions with government and industry representatives centered on ideas for partnering to build satellites, which are discussed herein. However, it must be emphasized that partnerships beyond just satellite constellations are important in space security mission areas. For example, the missile detection and tracking mission must operate on militarily relevant timelines, which demands equal attention to the ground architecture, data processing and analysis, and connectivity to missile defense systems. There is just as much opportunity to forge partnerships around ground architecture activities as there is in satellite development and production.
- **Complementary industrial base strengths:** Both nations emphasize the importance of industrial partnerships for delivering military space capabilities within relevant timeframes. However, there will be a natural tension as each country seeks to expand its industrial base and market access and decide on what each is willing to rely on the other for. Each

nation's industrial base retains unique and complementary strengths. For example, while U.S. industry is strong in systems integration, ground architectures, and large-scale satellite manufacturing capacity, Japanese industry is a global leader in satellite components, robotics, and advanced manufacturing.

- **Timelines, resources, and risks:** The pacing threats facing both nations drive aggressive timelines for fielding new capabilities. There may be near-term approaches that rely more heavily on U.S. industry or commercial service providers but offer longer-term pathways to increase Japan's own industrial capacity. Such a longer-term framing that articulates the overall security and economic benefits each country accrues will be important to both countries' stakeholders. Moreover, with the yen weakening, Japan's spending power will not go as far as anticipated despite significantly increasing investment in space security capabilities and personnel. On the U.S. side, defense budgets are likely to be flat or declining. Partnerships not only provide the benefit of increased capacity and resiliency for both countries, but also ways to burden share across resources and risk.
- **Security and digital resiliency:** There is now clear recognition across Japanese government, industry, and academia that security—cybersecurity, information security, personnel security, and facility security—is paramount to further U.S.-Japan space cooperation. Further, Tokyo is attuned to the importance of continuity of operations, particularly having observed Russian cyberattacks against Ukrainian satellite and ground networks and the effects of natural disasters on critical infrastructure such as the Fukushima nuclear power plant, which was damaged by a 2011 earthquake.

Recommendations

“The Japan-U.S. Alliance, including the provision of extended deterrence, will remain the cornerstone of Japan’s national security policy.”

– National Security Strategy of Japan, 2022¹²

This report advances several recommendations for deepening U.S.-Japan cooperation and collaboration in space security. The authors stress that collaboration should not be pursued merely for its own sake—there must be intent behind these actions to ensure the respective industrial, economic, and security needs are met for both nations. Summarized here and discussed in greater depth later in the report, these are areas where the combined, complementary strengths of both nations' governments and industries can be brought together to deliver concrete capabilities to advance both U.S. and Japanese national and collective security interests.

- **Cultivating human development:** While Japan grew into one of the world's most advanced spacefaring nations through its civil space program, its past limitation that space be used only for nonmilitary purposes resulted in its defense ecosystem having less experience in the use and development of space capabilities. Thus, building greater space proficiency within Japan's defense institutions and supporting industry through education and training will be foundational to any space cooperation activities.

- **Exercising and training the space force and joint force:** Japan should be encouraged, with support from the United States, to develop a greater understanding across its military services of how space capabilities contribute to security and defense missions by integrating space capabilities into joint exercises and training. Within the military space community, participation in tabletop exercises and wargames, as well as greater hands-on operational training, aid in understanding space threats, operations, and decisionmaking processes.
- **Sharing data and information:** Sharing space and cyber threat assessments is a logical start. Further, space domain awareness (SDA) data sharing can be enhanced and expanded trilaterally, building on the U.S.-Japan-South Korea trilateral information exchange model used for data sharing on North Korean missile launches.
- **Emphasizing complementary industrial base strengths:** In formulating partnerships, the United States and Japan should consider the unique and complementary strengths of each nation's industrial base and not limit their partnerships to the space domain. Both nations should look to each other as they seek to increase supply chain resiliency and seek investors to boost their space start-up ecosystems to achieve broader alliance objectives vis-à-vis space.
- **Architecting national security space constellations:** Both countries have set out national security space architectures that are ambitious, resource-intensive endeavors, with partnerships being a core tenet of their respective approaches. The areas of missile detection and tracking, ISRT, SDA, and satellite communications (SATCOM) stand out because of the urgency described in Japan's strategy documents to have sufficient defensive capabilities deployed by FY 2027. In several of these areas, accessing commercial space-based data and services can accelerate the delivery of near-term capabilities and complement government owned and operated architectures.
- **Investing in digital transformation and cybersecurity:** The United States and Japan should include a digital transformation and cybersecurity component within the bilateral space agenda. A digital infrastructure that encompasses data architecture, cloud computing, and cybersecurity can be a catalyst for greater interoperability and information sharing at an operational level.
- **Leading on policy and in international forums:** Through complementary and coordinated efforts in regional and international forums, both Japan and the United States can provide strong thought leadership and advance diplomatic initiatives on space sustainability, the development of responsible behaviors in space, and the reduction of threats to the peaceful uses of space.

While these recommendations can deliver material capabilities and policy successes, they will still require that policymakers address existing barriers. Chief among these obstacles is the need for leaders from both countries to set clear priorities and direction to move their respective institutions and industries into action. Both countries are undertaking ambitious space architecture plans. Delivering on those plans will be challenged over the long term by resource availability, industrial capacity, and labor availability in highly technical fields—all the more reason to forge industry relationships that leverage each nation's strengths.

Further barriers include developing a data sharing policy within the United States that provides guidance on the technologies, capabilities, and mission knowledge that U.S. industry can share with its Japanese counterpart, as well as Tokyo implementing security improvements, including legislation on security clearances extending from the military to industry, which it has made progress on. U.S. security experts across government and the private sector could be brought to Japan to share best practices.

Both countries' export policies and regulations have been barriers to closer space cooperation and industry-to-industry partnerships. U.S. International Traffic in Arms Regulations (ITAR) hampers bilateral industry cooperation, with both U.S. and Japanese companies which have been encouraged to forge joint space acquisition partnerships citing ITAR and classification as issues.¹³ Notably, in late 2023, Tokyo revised its principles on the transfer of defense equipment and technology, which is expected to enable greater export of domestically produced military technologies and components.¹⁴ While a positive development that can further contribute to supply chain diversification and the defense of like-minded partners that are increasingly under threat, the impact of this policy shift for space-related technology exports is unclear.

Finally, a consistent message from government and industry representatives in both countries is that they often do not know who to talk to on the other side. Both sides have found the annual comprehensive space dialogue to be highly valuable in bringing together both countries' space leaders and identifying counterparts, as well as facilitating interagency dialogues within each country. Opportunities to sustain these interagency and bilateral interactions beyond the senior leader dialogue were encouraged.

Although this study incorporates months of research, detailed briefings, and extensive discussions with government officials, industry representatives, and academic experts in both Japan and the United States, the authors acknowledge that it is not a comprehensive road map to all the opportunities and challenges in space security cooperation. However, as much as possible, the authors sought to identify concrete actions, including near-term options, that policymakers could take that, over time, would lead to material improvements in capabilities, more fielded systems, and a more highly trained and proficient workforce.

Introduction

Today, Japan sits in the middle of a complex Indo-Pacific security environment. Intensifying Chinese and North Korean military activities dominate regional security concerns, while Russia's invasion of Ukraine has provided a sobering wake-up call for what aggression in the region could look like. These concerns are reflected in three strategic security documents that Tokyo released in 2022: the National Security Strategy (NSS), National Defense Strategy (NDS), and Defense Buildup Program. These documents emphasize a need for independent security and economic growth at an accelerated pace to protect against growing threats, the risk of invasion from a neighboring nation, and instability around the world. Integrated into each of these three documents is the use of space (or *uchū* in Japanese) for security.

After World War II, the Allied powers imposed restrictions on Japan's heavy industry and defense capabilities, including restrictions on aerospace development. This continued until the San Francisco Peace Treaty of 1951, which ended the legal state of war and reestablished peaceful relations between the Allied powers and Japan through the United Nations.¹⁵ The treaty removed several restrictions, including those preventing Japan from entering the space age. Among the restrictions that were lifted, the treaty allowed for Japanese entry into the space age. In 1955, Professor Itokawa Hideo, known as the father of Japanese rocketry, launched a miniature nine-inch rocket, known as the Pencil Rocket, with a team from the Institute of Industrial Science at the University of Tokyo, and thus began Japan's ambitions in the space domain.¹⁶

For the decades that followed, Japan's national policy allowed space development only for peaceful purposes (which Japan defined until 2008 as excluding military space purposes), leading to Tokyo's

decades-long focus on research and scientific development in space, led by its civilian space organization, JAXA, and JAXA's predecessor organizations. However, a fraught security environment in the Indo-Pacific region has prompted Tokyo to reevaluate its security and economic goals. As the country reckons with an increasingly tense security dynamic driven by growing Chinese and North Korean military activity and defense budgets, space capabilities are becoming integral to Japan's national security. Further, the U.S.-Japan security alliance continues to be at the core of both countries' approaches to security and stability in the Indo-Pacific, with both seeking to deepen their cooperation across myriad national security areas, including in the space domain. Space is now a prominent agenda item in meetings between the U.S. president and Japanese prime minister, as well as several of their cabinet secretaries.

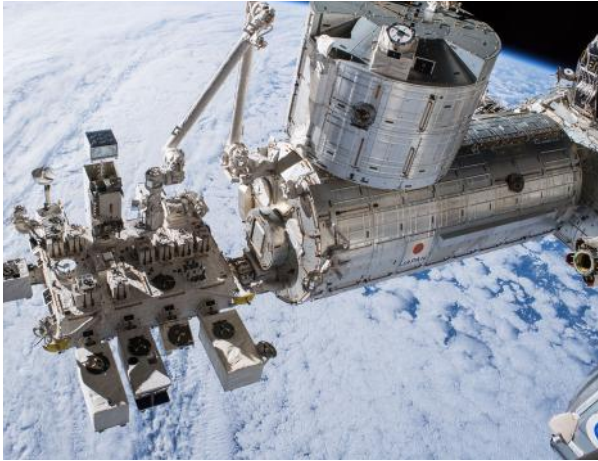
This report summarizes an Aerospace Security Project research effort to examine the evolution of Japanese strategy and policy for space security and identify potential areas for space security cooperation between Japan and the United States, including opportunities for industry engagement. The authors reviewed publicly available information; conducted interviews and discussions with a wide range of Japanese and U.S. officials, both in the United States and Japan; and hosted a series of workshops at CSIS with current and former Japanese and U.S. government officials, industry representatives, and academics. Valuable insights were gained regarding the trends, capabilities, priorities, strengths, and challenges within each country's space ecosystems.

This report outlines the history of space programs in Japan and the country's space capabilities, developments in space security after the 2008 Basic Space Law, international partnerships, regional space security observations, and alignment with the United States. It continues with a discussion of candidate areas to deepen or expand U.S.-Japan space security cooperation, including across missions and architectures, organizations, industrial and technology sectors, and within international forums. It also discusses potential barriers and limitations that need to be overcome to achieve greater cooperation.

Space History of Japan

Three distinct but cooperative organizations were tasked with incubating Japan's nascent national space program: the National Aerospace Laboratory of Japan (NAL), the Institute of Space and Aeronautical Science (ISAS), and the National Space Development Agency (NASDA). NAL was created in 1955 to research aerospace transportation systems.¹⁷ ISAS was founded in 1964 by the University of Tokyo for research and development and, in 1970, was responsible for the launch of the country's first satellite. In 1969, NASDA was founded by national law and set space development priorities for Japan.¹⁸ The law that created NASDA was accompanied by a supplementary resolution that limited the development and use of space to peaceful purposes.¹⁹ This resolution was interpreted by the Japanese government as "nonmilitary"; while many other countries around the globe also refer to "peaceful purposes" vis-à-vis their space activities, they often interpret the term to mean nonaggressive but still permissive of military space activity.²⁰

NAL, ISAS, and NASDA each held responsibility for different aspects of the country's space ambitions until they were later combined to create JAXA, which was established by the Law Concerning Japan Aerospace Exploration Agency in October 2003.²¹ Under this law, JAXA's objectives are to (1) develop research at academic institutions and universities, (2) enhance space and aeronautics science and technology, and (3) promote the utilization and development of space, but only for peaceful purposes.²² JAXA has continued to focus on peaceful operations, although it was authorized in the early 2010s to engage in security-related activities. It has since signed agreements with Japan's Ministry of Defense (MOD) for technical cooperation.²³



JAXA's Kibo laboratory on the International Space Station.

Photo: NASA

International Space Station (ISS) for microgravity experiments.²⁷ In 2022, Japan officially signed on to contribute to the ISS—which had been in operation for over 23 years—through 2030, making Japan the first country besides the United States to formalize its commitment for an additional eight years of operation.²⁸

Although JAXA's primary focus is on civil missions, Japanese space policies describe a partnership between JAXA and Japanese national security entities in developing and deploying certain space missions and technologies. For example, because of increased missile activity from North Korea in the late 1990s, including a North Korean rocket that flew over Japan in 1998, Japan launched its first satellite constellation, called the Intelligence-Gathering Satellite (IGS) program, in 2003.²⁹ IGS satellites are dual use between national defense applications and civil national disaster monitoring. These reconnaissance satellites have been launched with either optical (EO) or synthetic aperture radar (SAR) sensors, and each satellite has a projected operational life of five years on orbit.³⁰ The IGS program has seen over one dozen launches atop JAXA rockets in its 20-year history to include both EO and SAR satellites. However, the data from the IGS program is managed almost exclusively by the Cabinet Satellite Intelligence Center (CSICE).³¹ Japanese policymakers have sought to increase JAXA's contributions to Japanese space security efforts following the passage of the 2008 Basic Space Law, and collaboration is ongoing on space situational awareness and other areas, although there remains room for significant expansion of such efforts. JAXA is also working with Japanese industry to develop reusable rockets by 2030 to reduce launch costs and compete with international efforts to do the same.³²

In the 20 years since its founding, JAXA has become a reliable actor in civil space missions. One of the first missions after the organization's establishment was the Hayabusa mission, launched in May 2004, which collected matter from the near-Earth Itokawa asteroid for further study.²⁴ This mission made Japan the first nation to collect and return asteroid samples to Earth.²⁵ The Hayabusa mission was successfully completed in 2010 and a second mission, Hayabusa2, was launched in 2014 to collect samples from the Ryugu asteroid. Hayabusa2 successfully delivered samples back to Earth in 2020.²⁶

In 2008, JAXA contributed the Japanese Experiment Module, known as Kibo, to the

Space Organizations of Japan

One challenge to understanding the leadership and locus of space policy in the Japanese government is the siloed nature of space across sections of the Japanese government. In the executive branch, the prime minister is supported by the cabinet, which includes the Cabinet Office, Cabinet Secretariat, and 11 ministries. Space activities are ultimately overseen by the Strategic Headquarters for National Space Policy. As space is further integrated into defense, the Japanese MOD has become an increasingly important space stakeholder as well. Interministerial rivalry on space issues is common in Japan, and space activities often end up siloed, with a lack of formal coordination across ministries, until issues reach the National Space Policy Secretariat or the Strategic Headquarters.

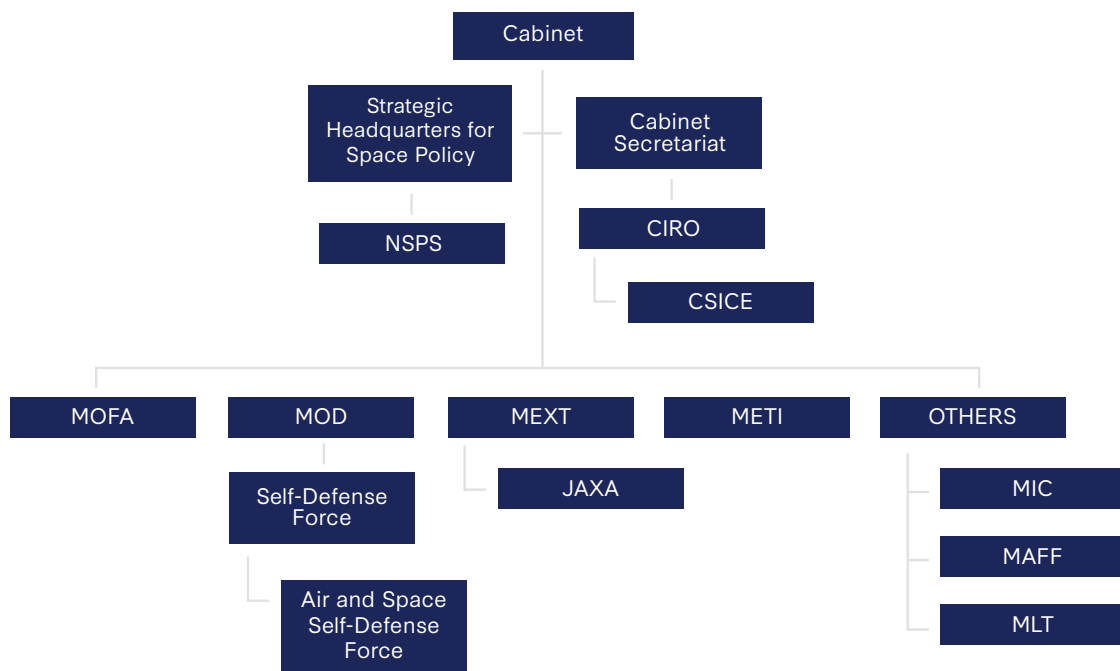
In the cabinet, a Strategic Headquarters for National Space Policy was created in 2008 by the Basic Space Law.³³ Under the headquarters, Japan's National Space Policy Secretariat (NSPS) was created in 2012 to assist in the coordination of space policies and budget allocations among the various ministries. It is considered a counterpart to the U.S. National Space Council (NSpC) staff. Similarly, the Committee on National Space Policy, involving representatives from industry and academia, was created in 2012 to serve as an advisory body to the headquarters. In contrast to the U.S. NSpC, the NSPS also holds licensing responsibilities under Japan's Space Activities Act and Remote Sensing Data Act.³⁴

The Cabinet Office, which operates under the cabinet, directs the planning, design, and coordination of policies regarding space exploration and the development, maintenance, and operation of the Quasi-Zenith Satellite System (QZSS). QZSS is Japan's regional satellite navigation constellation that both operates independently and is interoperable with the Global Positioning System (GPS).³⁵ The cabinet directed the Cabinet Office to operate QZSS.

The ministries with primary responsibilities for Japan's space programs are the following:

1. The Ministry of Education, Culture, Sports, Science, and Technology (MEXT), which oversees JAXA
2. The MOD, which oversees the Japanese military and Self-Defense Force (SDF), as well as the recently established Space Operations Group³⁶
3. The Ministry of Economics, Trade Industry (METI), which has become more involved with space and helps administer strategic planning and industry funds³⁷
4. The Ministry of Foreign Affairs (MOFA), which handles international space policy diplomacy
5. The Ministry of Internal Affairs and Communications (MIC), which oversees the National Institute of Information and Communications Technology and its large space research and development group
6. The Ministry of Land, Infrastructure, Transport, and Tourism (MLIT), which oversees the Japan Meteorological Agency, responsible for the Himawari geostationary meteorological satellite constellation
7. The Ministry of the Environment (MOE), which is responsible for monitoring greenhouse gasses
8. The Ministry of Agriculture, Forestry and Fisheries (MAFF), which is responsible for supporting relevant industries and food security

Figure 1: Japan Space Organizations



Source: Authors' analysis.

Space responsibilities are also assigned to the Cabinet Intelligence and Research Office (CIRO). Although not part of a ministry, CIRO falls directly under the prime minister and is responsible for providing integrated intelligence to policymakers in the cabinet and the Cabinet Secretariat. Within CIRO, the Cabinet Satellite Intelligence Center is responsible for acquisition, operations, data collection, and imagery analysis for the growing IGS satellite constellation.³⁸

Ministry of Defense

The MOD organizes and executes national defense strategy and leads the Self-Defense Force. Space policy and program decisions are created throughout a variety of positions and offices in the MOD, specifically including the Internal Bureau of Defense Policy and Bureau of Defense Buildup Planning. Other space-related organizations include the Joint Staff; Air Staff Office; the Acquisition, Technology & Logistics Agency; and the Defense Intelligence Headquarters.³⁹ Japan is set to stand up the Japan Joint Operations Command by 2025, which will direct all SDF joint operations.⁴⁰

The military service responsible for space activities is the Japan Air Self-Defense Force (JASDF), which was established in 1954. In FY 2027, the service will be renamed the Japan Air and Space Self-Defense Force to “reinforc[e] its space operation capability and . . . [develop] a system to ensure superiority in use of space.”⁴¹ The Space Operations Group was created in 2020 as the first space-focused unit within the Japan Air Self-Defense Force.⁴² The group is organized to facilitate easy cooperation and interoperability with parallel U.S. organizations. It currently has 150 military personnel and will continue to grow.⁴³ Its initial responsibility is the operation of Japan’s Space Situational Awareness (SSA) system. The group has four main objectives: (1) enhancing SSA; (2) improving information, communication, and positioning capabilities; (3) leveraging electromagnetic domain capabilities to deny other actors use of the full slate of command, control, communication, computer, and intelligence operations; and (4) working with other Japanese agencies, such as JAXA, and relevant counterparts in the United States and around the world.⁴⁴ Beyond the Space Operation Group, the JASDF chief of staff maintains relationships with space leaders around the world.⁴⁵

Evolution of Space Security Policy in Japan

Increased ballistic missile and space activities from China and North Korea began to shift the security dynamic in the Indo-Pacific and motivated Japanese leaders to rethink Tokyo's national defense strategy. Japan began to slowly integrate space into its overarching security policies starting in the late 2000s. China's 2007 direct-ascent anti-satellite (ASAT) test, which created thousands of pieces of debris in low Earth orbit (LEO), also likely influenced Japanese perspectives on space. Japan was one of the first to condemn the test, along with the United States and the United Kingdom, and many others followed.⁴⁶

In 2008, one year after China's destructive ASAT test, the Diet adopted the Basic Space Law, which went through many iterations to gather support from both the ruling and opposition parties in the Diet.⁴⁷ The Basic Space Law was the first law for space development in the country and reversed Japan's decades-long policy that space be utilized for nonmilitary purposes only, aligning Japan with the same policy approach as many other countries. It established a Strategic Headquarters for National Space Policy, which became a driver of space policy and plans in the following years.⁴⁸

Basic Space Plan

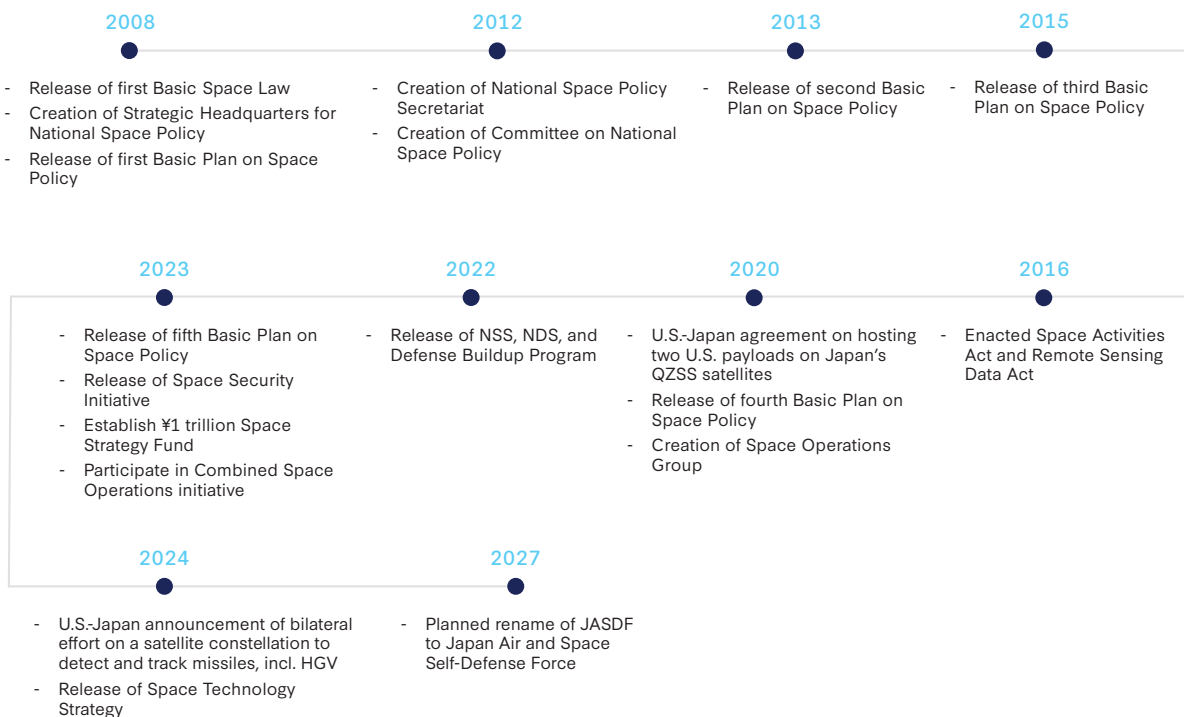
The Basic Space Law also created the first Basic Plan on Space Policy, which was a foundational document in Japanese space policy and the first to clearly identify national security as an objective of Japanese space activities. This was a turning point in how Japan approached space capabilities as part of its national defense strategy. The Basic Plan on Space Policy is now in its fifth iteration.⁴⁹ In both the Basic Space Law and the iterations of the Basic Plan on Space Policy that followed, the

development and use of space capabilities are described as a way to improve the lives of Japanese citizens and increase international peace and security.⁵⁰

In January 2013, Japan published a second version of its Basic Plan on Space Policy that outlined government-led space activities for the decade ahead, citing a rapidly changing space domain as an impetus to update the country's policies. The document suggested a fundamental shift in Japanese space development from a focus on scientific and technological research to the utilization of space for national security and environmental monitoring. Further, it emphasized the need for Japan's aerospace industry to become a competitive supplier of space capabilities and services to buyers outside of Japan in an increasingly international market.⁵¹

Two years later, in January 2015, a third Basic Plan on Space Policy was published, with a request that two industry-focused space acts be adopted by the Diet.⁵² These two laws—the Space Activities Act (SAA) and the Remote Sensing Data Act—were enacted in 2016.⁵³ The SAA was created to establish a system to regulate and license the launch of spacecraft, ensure public safety, and promote the use of space to improve the lives of the Japanese people, including via economic development.⁵⁴ The Remote Sensing Data Act is a broadly scoped document that established a licensing regime for space-based remote sensing operations.⁵⁵

Figure 2: Japan's Space Security Advancements since 2008



Source: Authors' analysis.

In June 2020, the country released its fourth Basic Plan on Space Policy, acknowledging for the first time the urgency of space assets to advancing national security. This update describes Japan's space industry as lagging behind other countries in trending areas, such as the deployment of large constellations of small satellites. The document describes the growing importance of space in national security, the dependence of militaries and governments on space systems, and the risks of interfering with the sustainable use of the space domain. It also lays out Japan's space policy vision: to become a self-sustaining space nation and use space to contribute to a wide variety of national interests, including disaster management and economic growth. An undercurrent of the document is a desire to use space to solve global sustainability issues and strengthen national resilience.⁵⁶ Sustainability and resource management are two key pillars of Japanese society and policy and remain a priority for space development.

A fifth Basic Plan on Space Policy was released in June 2023, highlighting the accelerated use of space for national security and the growing dependence of economic and social activities on space systems.⁵⁷ The document further notes that on-orbit systems and their ground architectures are increasingly becoming seamlessly integrated and that access to space is democratizing beyond traditional spacefaring nations, which the document labels "space transformation."⁵⁸ The plan further underscores that the symbiotic nature between private and government space investment helps Japan maintain its leading role among spacefaring nations and must be further strengthened.⁵⁹

2022 Security and Defense Strategies

Japan released three new strategic documents as well as an updated defense budget document in December 2022. The NSS, NDS, and Defense Buildup Program together reflect a turning point in Japan's defense positioning.⁶⁰ Integrated into each of these documents is a strong emphasis on the use of space for defense and national security. This is a noticeable change from policies of the past.⁶¹ Japan does not yet operate any satellites exclusively for military purposes, consistent with previous policy, though the SDF exclusively operates military-use communications payloads on private sector satellite buses.⁶²

NATIONAL SECURITY STRATEGY

The complex and increasingly threatening security environment frames the 2022 NSS. Described as the most "severe and complex as it has ever been since the end of World War II," the security environment includes the expanding military arsenals of China and North Korea, the introduction by China and North Korea of advanced weapons such as HGVs, and Russia's brazen invasion of Ukraine and Ukraine's effective resistance.⁶³ The NSS draws parallels between Russia's hostile actions against Ukraine and the West and potential Chinese regional aggression against Taiwan and others. The NSS also clearly and frequently mentions the threat of a nuclear weapons buildup in China, North Korea, and Russia.⁶⁴

As a result of growing security demands, these new documents emphasize the need for an independent defense strategy and accelerated economic growth to protect against threats and the risk of invasion from a neighboring nation, as well as challenges to the post-World War II international order.

The NSS, which was first issued in 2013, outlines three main national interests: (1) sovereignty and independence while maintaining peace and security; (2) economic prosperity; and (3) protecting universal values and the international order, specifically in the Indo-Pacific region. It also calls upon like-minded nations to come together and cooperate to address global challenges, such as climate change and infectious diseases. The document also states that the line between military and nonmilitary fields has been blurred in areas both technological and economical, another departure from previous national documents that explicitly separated the two.

Space is cited heavily in the strategy and is specifically mentioned as a national security policy area, alongside diplomacy, defense, and economic security. The NSS calls for increased comprehensive efforts in space security from the entire government, including the use of the space domain by the Japanese Self-Defense Forces and the integration of space, cyberspace, and electromagnetic domains into ground, maritime, and air capabilities.⁶⁵

Missions prioritized in the NSS, including both missile defense and counterstrike operations, will rely heavily on space-based capabilities. The Japanese government had not previously acquired counterstrike capabilities due to the interpretation of its “defensive defense” policy at the time. As the Japanese military is still a self-defense force, government policy permits counterstrikes, but preemptive strikes are not permitted—a distinction that comes after years of debate in the Japanese government dating back to the 1950s. Currently, Japan relies significantly on the United States for information to enable its own missile defense and counterstrike capabilities. The expanded threat of mobile missiles, with unusual trajectories and the ability to be launched en masse, along with the capabilities of hypersonic weapons, are heavily underscored in the document and a key driver for the need for information collected from space-based assets. This includes imagery satellites to detect missile movements on the ground, develop targeting solutions, and provide warning of impending launches, as well as satellites that can detect missiles once launched and develop tracks and firing solutions for missile defense interceptors.

For the first time in a Japanese national security strategy document, space security has its own subsection, “Reinforcing Comprehensive Efforts for Space Security,” in which space is described as “indispensable” for social and economic activities.⁶⁶ The SDF and Japan Coast Guard are encouraged to utilize space capabilities. The section also calls for strengthening cooperation between JAXA and SDF on security matters—language that is a departure from past space policies in which JAXA was isolated from defense and security discussions. Greater “cooperation and coordination” are also encouraged between the CSICE, MOD, and SDF in intelligence gathering, including from space-based assets.⁶⁷ Furthermore, the section encourages the utilization of civilian space technology and growth in the space industrial base to support the development of satellite constellations and better integration of industry products into space security missions.

The document itself does not lay out specifications for space priorities or space organizations, but it sets the expectation that the country will release a concept supported by the government. This Space Security Initiative—the country’s first space security policy—was announced in June 2023 at a meeting chaired by Prime Minister Kishida Fumio.

Finally, the importance of the U.S.-Japan relationship in the NSS cannot be understated, nor can the emphasis on deepening security cooperation between the two countries. As Japan's ally "with the world's greatest comprehensive power," increased cooperation and coordination with the United States is listed numerous times as a strategic approach and a means by which to achieve Tokyo's security goals, including in the space domain.⁶⁸

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NATIONAL DEFENSE STRATEGY

The 2022 NDS seeks to clarify the national defense objectives of Japan, maintaining that the development of defensive capabilities is solely to protect the nation, not to pose a threat to other countries.⁶⁹ Building off the NSS, the NDS is framed by Chinese, North Korean, and Russian threats to both the national security of Japan and the international community. The NDS articulates that the Indo-Pacific region is facing significant challenges:

- China is building up its military and technological resources to amplify force on its mainland and in the East China and South China Seas.
- North Korea has become more brazen with its ballistic missile programs and pursuit of nuclear weapons.
- Russia's 2022 invasion of Ukraine and its impacts in Europe are a warning to the possibilities of invasion or disruption in the Indo-Pacific.⁷⁰

The document calls for a strengthening of Japan's defense capabilities and overall deterrence posture, particularly in cooperation with the United States, whose alliance serves as a pillar of Tokyo's overall security policy.⁷¹ Time is a grounding factor in the NDS, which calls for Japan to be able to primarily defend itself against potential invaders by 2027.⁷²

Although the use of space for missile warning is not explicitly stated, missile defense is a stand-out priority throughout the document. Missile defense, and a subsequent counterstrike capability, have rising importance in light of elevated missile launch activity from North Korea and China inching closer to Japanese territory. Space is the key enabler for intelligence, surveillance, reconnaissance, and targeting capabilities to ensure accurate and responsive missile warning and defense in the nation. Japanese defense policymakers are prioritizing Japan developing its own capability to defend itself, with missile defense systems developed in collaboration with and built to be interoperable with U.S. capabilities.

The NDS also details a desire to integrate space into MOD and SDF activities by utilizing satellite constellations for communications, intelligence gathering, and positioning and by building a space domain awareness (SDA) structure to respond to threats in the space domain. Finally, the

NDS calls for an enhanced SDA capability to be able to monitor objects in space by FY 2027 and for the continued acquisition of space capabilities and integration into SDF units over the next 10 years.⁷³

Similar to the NSS, the NDS recognizes space as vital for both civilians and the military and calls for increased collaboration between the SDF and JAXA.⁷⁴ The NDS also renames the Air Self-Defense Force as the Air and Space Self-Defense Force, emphasizing a renewed focus on space.⁷⁵

DEFENSE BUILDUP PROGRAM

The Defense Buildup Program serves as a guide for the MOD and the SDF to build, maintain, and operate defense capabilities.⁷⁶ It provides a general outline of the systems the SDF will procure and operate. The program states that within five years, Japan will strengthen its defense capabilities, and within ten years, the SDF will be capable of deterring future invasions without allied assistance. Space capabilities are called upon early in the document as a tool to improve cross-domain operations such as missile warning, information gathering, and communications.⁷⁷ The SDF is directed to develop a space-based space domain awareness capability, which will include the use of supplementary commercial options, and to increase efforts to add layers and resiliency to existing secure military communications. The MOD will also coordinate with JAXA to leverage space as a mechanism to strengthen cooperation with the United States and other friendly nations. Finally, the program echoes the call of the NDS to rename the country's Air Self-Defense Force to the Air and Space Self-Defense Force, highlighting the increased role space capabilities will play in the military.⁷⁸

Throughout each of these documents, it is clear that space has become an integral component of Japan's national security posture. With a focus on information gathering, communication, and positioning data from space capabilities, Japan is focused on further integrating space, including JAXA and industry partners, into its defense architecture.

2023 Space Security Initiative

To build on the NSS, Japan released the Space Security Initiative (SSI) in June 2023 to provide a decade-long strategy for space security and the space architecture needed to achieve those goals. The initiative identifies space as an “arena for geopolitical competition for national power over diplomacy, defense, economic[s], and intelligence. . .” and highlights a growing need for an expansion of capability and cooperation with allies.⁷⁹ The document outlines requirements for a national security space architecture to include compatibility and interoperability of satellite data, cyber security, and information security; resilience against threats and risks in space; and economic efficiency through the use of commercial services.

The initiative identifies parallel framing issues as in the NSS, but it is able to specify the use of space in target areas of policy. The SSI describes the impact Western commercial satellites made in supporting Ukraine in the initial weeks of Russia's invasion, describes the critical importance of cloud-based data management, and references China's dramatic increase of space capabilities to include SAR and radio frequency satellites.⁸⁰

The SSI describes a three-pronged approach to establishing a national security space architecture: “Radically Expand the Use of Space Systems for National Security,” “Ensuring Safe and Stable Use of Outer Space,” and developing a “Virtuous Cycle of Security and Fostering [a] Space Industrial Base.”⁸¹ A thorough and ambitious architecture is presented in the document, with space capabilities covering position, navigation, and timing (QZSS and GPS); intelligence (IGS constellation and commercial imagery satellites); communications (commercial SATCOM, defense SATCOM, tactical SATCOM, commercial and optical constellations); missile defense (early warning satellites and a missile defense space system); space domain awareness (SDA satellites and commercial space situational awareness satellites); and on-orbit fuel supply satellites. It also includes space transportation (government launch vehicles and commercial rockets) and optical telescopes and radar run by JAXA.⁸²

The SSI outlines specific needs for a national security space architecture. Satellite data must be interoperable with U.S. and allied infrastructure. The architecture must have robust cyber and information security; be resilient against space threats, including anti-satellite weapons; be able to protect ground stations; and be able to leverage commercial services, particularly in communications, Earth observation, and data analysis.

To achieve security from space, the first priority listed is intelligence collection. This is specified as geostationary-based optical satellites to add to the current IGS constellation. The current IGS system, and intelligence collection systems writ large, place resolution above revisit rate and are stovepiped from other MOD initiatives, creating the need to duplicate particular systems to give information to disconnected government offices. The SSI calls for an increase in satellite resolution, faster data transfer, and cooperation among CSICE, which currently operates and owns IGS data, the MOD, and the Japanese Self-Defense Forces, as well as other relevant agencies.

The second priority outlined in the SSI is missile defense, specified as continued work with the United States on an early warning satellite. Additionally, the SSI describes a technical demonstration of additional missile defense capabilities, such as detection and tracking to respond to ballistic missiles and HGVs being developed and deployed by China and North Korea. This may be the architecture that presents the most opportunity for collaboration and coordination with the United States.

The initiative’s third priority is the establishment of anti-jamming communications capabilities. Tokyo is attuned to the importance of continuity of operations, seeing Russian cyberattacks against Ukrainian communication networks on the first day of the conflict. The SSI calls for a layered and redundant communications capability that utilizes multiple orbits, both geostationary orbit (GEO) and LEO, to be interoperable with the United States and additional allies. This communications system will be used not only by the MOD and Self-Defense Forces, but also by the Japan Coast Guard.

The fourth priority relates to position, navigation, and timing services, to be achieved through strengthening the current QZSS constellation. The final aspect of security from space in the SSI is

space transportation, with a focus on continuing to operate a space launch vehicle, reduce launch costs, and commercialize operations for government utilization.

The SSI then turns to priorities for security in space and ensuring the safe and sustainable use of the space environment. The top priority in this initiative is enhancing space domain awareness. This encompasses not only situational awareness and movements of other satellites, but also operations and capabilities of other nations' on-orbit assets. Space domain awareness will offer avenues to collaborate internationally in the Combined Space Operations Center and the Combined Space Operations Initiative.

On-orbit servicing is a secondary priority area, focusing on on-orbit refueling and servicing to extend the life cycle of satellites in GEO. Subsequently, the SSI details the importance of information sharing across Japanese government agencies that may currently be stovepiped in the decisionmaking process, including the Cabinet Secretariat, the Cabinet Office, the MOD, and the Self-Defense Forces. Strengthening cooperation and information sharing with the United States is highlighted throughout.

Finally, the SSI specifies that Japan will continue to lend its voice to the United Nations and other multinational organizations to shape norms for responsible behavior in the use of space. Along with the United States, Japan has committed to not conducting destructive, direct-ascent ASAT missile testing and to mitigating any future creation of space debris.

To realize its ambitious goals, the SSI acknowledges the need to strengthen public-private partnerships in the space domain in a large variety of areas, including personnel exchanges and technical cooperation with industry and government both within and outside Japan; research and development; government support of industry; investment diversification in private entities; and additional backing to industries with dual-use opportunities such as on orbit servicing, SAR and optical satellites, and launch. Additionally, the SSI aims to support a stable supply chain for components of critical technologies, including by issuing a Space Technology Strategy and by bolstering JAXA as the center of space excellence in the nation.

The Space Technology Strategy was released in 2024 to build on the 2023 Basic Space Plan and SSI. It serves as a guide for the timeline and development of civilian and military sectors in satellites, space science and exploration, space transportation, and cross-domain technologies.⁸³ The Space Technology Strategy further prioritizes efforts to mitigate “supply chain risks” to Japan’s “predominance” in space technologies, again linking the “competitiveness” of Japan’s space industry to its national security.⁸⁴

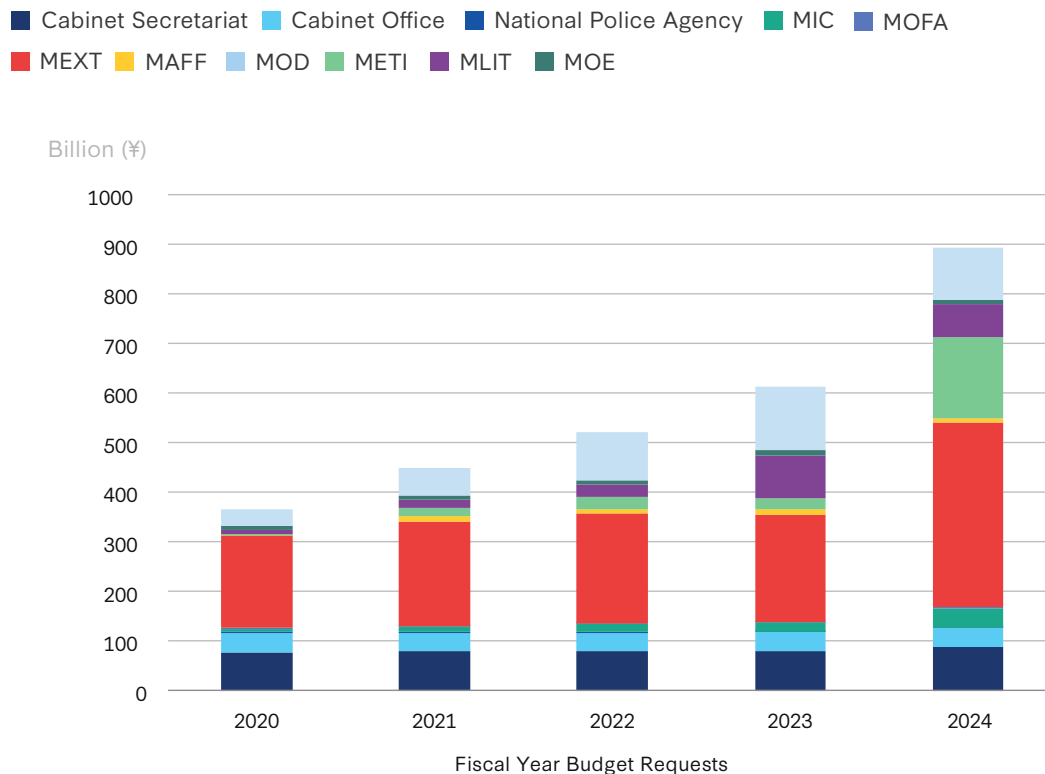
To support these priorities, in November 2023, Tokyo announced that JAXA would be allocated ¥1 trillion over a 10-year period to support the nation’s commercial sector and technology development in the space industry. The funding will focus on satellites, space exploration, and space transportation. An additional ¥1 trillion is anticipated to be allocated for space security from 2023 to 2027.⁸⁵ Like the United States and others, Japan recognizes the economic and technological benefits of space and how its investments create a demand signal to spur private sector innovation

and attract technical talent. It also sees advances in the space domain helping solve global sustainability, resource management, and disaster relief challenges here on Earth.

Space Budget

Tokyo’s policies and plans for greater space utilization are being implemented through increased funding. According to annual budget request documents published by the Cabinet Office, the government of Japan has been steadily increasing its space budget and, since FY 2019, spending on space activities has more than doubled.⁸⁶ In late 2023, the government of Japan announced plans to commit ¥1 trillion (\$6.9 billion) over a 10-year period toward the expansion of Japan’s domestic space industry.⁸⁷ These resources, established as the Space Strategy Fund, will be managed by JAXA. The FY 2024 space budget begins to reflect this new funding, with large increases allocated to MEXT, the ministry overseeing JAXA, and METI, which has become more involved in stimulating Japan’s space industrial base.

Figure 3: Space Budget Across Japanese Ministries, 2020–2024



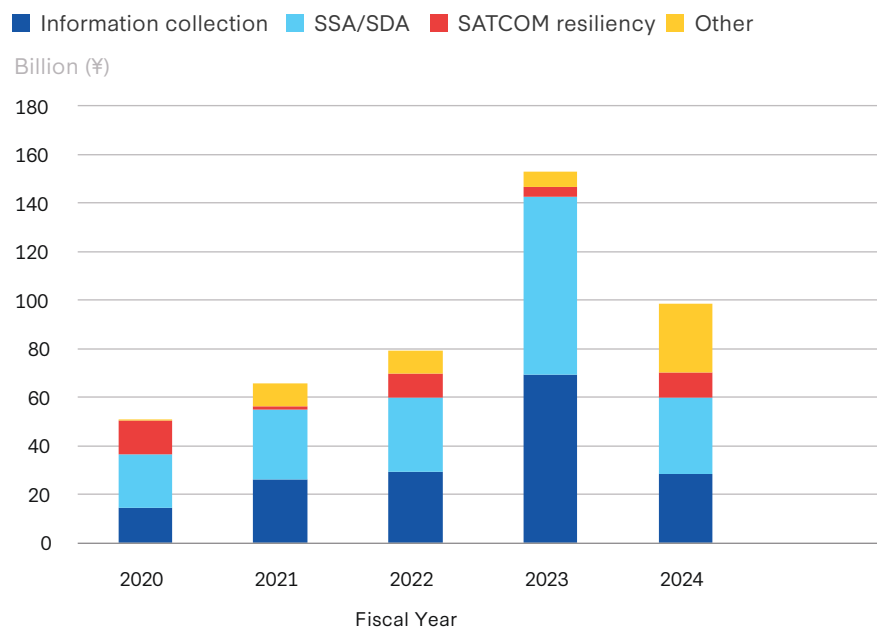
Source: “宇宙関係予算について” [Regarding the space-related budget], Cabinet Office, Government of Japan, <https://www8.cao.go.jp/space/budget/yosan.html>; and authors’ analysis.

Japan’s overall defense budget has been increasing since 2012, and it saw significant increases following the 2022 release of its defining defense strategy and budgetary documents. Tokyo’s FY 2022 defense budget (including its initial budget and supplementary budget)—framed as a “Defense-Strengthening Acceleration Package”—comprised ¥5.9 trillion (\$40.9 billion) and reflected

a 6.5 percent increase over FY 2021.⁸⁸ In FY 2023, Japan’s “First Year’ Budget for Fundamental Reinforcement of Defense Capabilities” totaled ¥6.6 trillion (\$45.7 billion), an increase of 11.9 percent over FY 2022. And in FY 2024, a year characterized as “Progress . . . in [the] Fundamental Reinforcement of Defense Capabilities,” Tokyo’s defense budget was ¥7.7 trillion (\$53.3 billion), an increase of 16.7 percent.⁸⁹ The FY 2025 budget request is planned for release in August 2024 and is expected to reflect continued increases in defense spending aligned with Tokyo’s national security and defense strategies.

The MOD publishes an annual overview of its budget for defense programs, with space domain capabilities categorized in a few broad areas: space situational awareness and space domain awareness, strengthening information-gathering capabilities, enhancing space resiliency, and utilizing space for missile defense.

Figure 4: Japan Ministry of Defense Space-Related Budget



Source: “Defense Budget,” Ministry of Defense, https://www.mod.go.jp/en/d_act/d_budget/index.html; and authors’ analysis.

Honing in on FY 2023, since that year marks the first MOD budget following the release of Japan’s NSS, NDS, and SSI, space funding nearly doubled, increasing from ¥79 billion (\$546 million) to ¥152.9 billion (\$1.1 billion).⁹⁰ The FY 2023 budget document framed this funding increase with the observation that “the space domain is now the foundation of our citizen[s]’ daily lives and security, and it is extremely important for Japan to ensure superiority in space use”; therefore, “it is necessary to strengthen space operational capabilities.”⁹¹ SDA enhancements saw the greatest growth, including the production of a domestic SDA satellite and consideration of more satellites (¥59.5 billion, \$411.9 million) and the development of space command and control capabilities (¥13.7 billion, \$94.8 million). The other area that saw significant growth was in information-collection capabilities, with ¥34.3 billion (\$237.2 million) allocated to enhancing the X-band SATCOM network

and ¥22.6 billion (\$156.2 million) allocated to the procurement of data for image analysis. Notably, the FY 2023 budget also begins funding infrared sensor technology demonstrations for HGV detection and tracking (¥4.6 billion, \$31.8 million) and demonstrations to acquire and process EO/SAR data and respond to HGVs (¥8 billion, \$55.3 million).⁹²

The FY 2024 MOD space budget slightly declined from FY 2023. It maintained an emphasis on SDA to include ¥17.2 billion (\$118.8 million) for the launch and deployment of an SDA satellite, scheduled for launch in FY 2026, and space command and control (¥9.2 billion, \$63.6 million).⁹³ The budget also begins funding an optical datalink demonstration in GEO (¥4.8 billion, \$33.2 million) and a study on hosted payloads. It continues investments in information-collection capabilities with the acquisition of commercial satellite imagery (¥24.7 billion, \$170.6 million) and advancing HGV detection and tracking technology demonstrations, specifying use of the HTV-X space station resupply vehicle as an experimental platform (¥3.8 billion, \$26.2 million). The one area that saw a significant increase over FY 2023 is in SATCOM resiliency to build interoperability between X-band military SATCOM and other commercial SATCOM systems (¥10.5 billion).⁹⁴

Experts on Japan's space programs and budgets shared with the authors that the FY 2024 topline decrease for space can be explained through the natural progression of certain programs (e.g., allocating funds to transition the SDA satellite from production to preparing it for launch and deployment) and some delays in budget execution. However, the FY 2025 space budget request, scheduled to be announced in August 2024, is expected to rise significantly.⁹⁵

Space Industrial Base of Japan

Japan's space industry is worth approximately ¥1.2 trillion (\$8.6 billion) as estimated by the World Economic Forum, with the Japanese government seeking to double the size of the industry by the early 2030s.⁹⁶ It is dominated by government-funded heavy industry, primarily through the production of space launch vehicles (SLVs), but includes a growing number of space start-ups. Japan's long-standing focus on indigenous launch capabilities and independent access to space indicates a prioritization on autonomy. SLVs are often seen by rising space powers as a key to competitiveness in the space market and international success in the domain.⁹⁷ A majority of SLV operations are now based in the private sector with JAXA oversight.⁹⁸ On the start-up front, over a 21-year period (2000-2021), Japan accounted for the third-largest number of investors in space start-up companies, behind the United States and China.⁹⁹

According to the 2019 annual report of the Society of Japanese Aerospace Companies, Japan's space industry comprises three broad categories: (1) space vehicles, including launch vehicles, satellites, and space station components, which make up 87 percent of total production; (2) ground facilities, which comprise 8 percent; and (3) software, which accounts for the final 5 percent.¹⁰⁰ The FY 2022 report, released in 2023, shows a slight rebalancing of 76.6 percent, 9.6 percent, and 13.8 percent, respectively.¹⁰¹

Traditional Industry

In 2019, the Japanese aerospace market was dominated by four large companies: Mitsubishi Heavy Industries (MHI), IHI Aerospace, Mitsubishi Electric Corporation (MELCO), and NEC Space



Japan's next-generation H3 rocket lifts off from the Tanegashima Space Center on July 1, 2024.

Photo: STR/JIJI Press/AFP via Getty Images

Technologies.¹⁰² These companies provide products and solutions across a wide range of industrial sectors—from agriculture to telecommunications to transportation. Aerospace and defense comprise a fraction of their overall business.

MHI has been providing launch services in Japan since 2007 and has proved to be a reliable partner to JAXA in the development of three SLVs: the H-IIA, H-IIB, and H3 launch vehicle (in production).¹⁰³ Prior to the H3's failure in March 2023, the H-IIA/B vehicles had 49 consecutive successful launches. In addition to SLV production, MHI produces rocket engines, test facilities, electronic parts, and parts of Japanese systems or components

for the ISS, such as the Japanese Experiment Module.¹⁰⁴ The H3 test-launched successfully in February and July 2024.¹⁰⁵

IHI is involved in a variety of large industries in Japan, including space, defense, aircraft engines, and long-term research and development. The Epsilon Launch Vehicle, developed by IHI Aerospace and operated by JAXA, is currently on its fifth iteration, known as Epsilon (S), and has a successful record of inserting up to nine satellites into LEO.¹⁰⁶ Additionally, IHI Aerospace contributes solid rocket boosters, a gas jet system, and pyrotechnics to the H-IIA and H-IIB launch vehicles and continues to work with JAXA and MHI on the production of a new solid booster for the H3 vehicle.¹⁰⁷ IHI Aerospace also designed and manufactured the return capsule for the Hayabusa asteroid probe.¹⁰⁸

MELCO has been developing space technology for Japan since the 1960s, with an emphasis on satellite production, high-performance payloads, and exportable components for satellite programs across the globe.¹⁰⁹ MELCO is also involved in building ground stations, including manufacturing antennas, transmitters, receivers, and radio and optical telescopes, among other products.¹¹⁰

NEC Space Technologies focuses on specific components for satellites and launch vehicles rather than full systems. Components from NEC have been used on over 350 spacecraft worldwide.¹¹¹ Though one of Japan's mainstay industry partners in space, NEC maintains fewer than 450 employees across its space business, serving as a reminder of the small scale of Japanese space personnel and the room for growth.¹¹²

SKY Perfect JSAT Corporation is the largest geostationary satellite communications operator in the Pacific region, where the company operates 17 geostationary communications satellites. The company was established in 1985 as Japan's first private SATCOM company. It plans on expanding operations to include Earth observation satellites to enable it to respond to security issues.

SKY Perfect JSAT is headquartered in Tokyo, with a U.S. subsidiary, JSAT International, which is headquartered in Washington, D.C.¹¹³

In addition to conglomerates that operate aerospace and defense segments, Japan has a unique trading house economy. Known as “sogo shosha,” Japanese investment houses, which are often part of larger Japanese business groups known as “keiretsu,” look to engage in a variety of sectors and technologies; they helped propel Japan into global markets and industrialization, particularly in the aftermath of the post-war reconstruction period. Modern-day houses are increasingly investing in space technologies and companies in Japan and internationally.¹¹⁴ Traditional firms are also investing in international space start-ups, including through joint ventures.¹¹⁵

New Space

Small start-ups, many of which are funded by venture capital, are expanding Japan’s space activities beyond the scope of traditional Japanese heavy industry.¹¹⁶ Several of these companies are capitalizing on Japan’s strength in robotics to innovate in the areas of space sustainability and exploration, such as space domain awareness and on-orbit servicing. Companies such as Astroscale, ispace, and GITAI are focusing on a variety of different mission areas, ranging from on-orbit servicing to lunar missions to robotics on the ISS. Other new space companies, including Axelspace, iQPS, and Synspective, are developing small satellite constellations to observe and analyze natural and human activity on Earth. Listed below are several new space companies, providing examples of the wide-ranging space technologies in development in Japan.

Astroscale, a Tokyo-based start-up company with subsidiaries in the United States and the United Kingdom, among other countries, specializes in on-orbit servicing, including satellite life extension, on-orbit SSA, and debris-removal technology for the safe and sustainable use of space.¹¹⁷ Although products from Astroscale are largely still in the development phase, the company has conducted successful test missions. In 2023, Astroscale was awarded a contract by the U.S. Space Force Space Systems Command (SSC) for the provision of on-orbit satellite refueling services, a demonstration of the potential for collaboration between the U.S. government and Japanese industry.¹¹⁸

Another innovative Japanese company is ispace, a lunar exploration company that also seeks to expand human presence beyond Earth.¹¹⁹ In late 2022, ispace launched its first Hakuto-R mission, with the goal of a lunar landing in early 2023. Although the lander successfully reached lunar orbit, it crashed into the surface of the Moon, likely due to a software failure, and was unable to complete its mission.¹²⁰ If it had been successful, the lander would have become the first private sector spacecraft to land on the Moon. Ispace’s second Hakuto-R mission, which will again attempt a soft landing on the Moon, is currently scheduled for 2024 launch.¹²¹ Additionally, ispace has been contracted by NASA to develop the lander for a 2026 commercial lunar payload services mission to the far side of the Moon.¹²² In late 2023, ispace established a U.S. headquarters in Colorado to enable more business opportunities in the United States, especially with the U.S. government.¹²³

GITAI is a space robotics start-up capitalizing on Japan’s long history of excellence in robotics. The company aims to partner with launch companies to build colonies on the Moon and Mars.

While this may sound far-fetched, the company has formed successful relationships and contracts with Toyota Motor Corporation and the Ministry of Economy, Trade and Industry; has completed a successful technology demonstration mission inside the ISS; and has engaged with U.S. space companies.¹²⁴ Like other Japanese space start-ups that have established a U.S. headquarters or subsidiary, GITAI has moved most business operations to the United States to better conform to U.S. export control regulations.¹²⁵

Axelspace designs and produces Earth observation microsatellites and their components and successfully launched its first satellite in 2014.¹²⁶ The company currently has five satellites in orbit with customers in over 30 countries. Axelspace's business model centers on sustainability, including using collected optical imagery and data services to monitor climate change resulting from deforestation and supporting a sustainable space environment through policy advocacy.

Similarly, iQPS and Synspective are building small satellite constellations for Earth observation, leveraging synthetic aperture radar (SAR) technology that enables the collection of data and imagery day and night, and in all-weather conditions. The information can be used to support environmental monitoring, disaster management, and national security applications.¹²⁷

Growing Prominence of Space in the Bilateral Relationship

“Our bilateral partnership is among the strongest in the world, in part because we’re always adapting it to meet the evolving challenges of our time. That’s the case on everything from advancing security and stability in the Indo-Pacific, to bolstering our economic and energy security . . . we’re also expanding the horizons of our partnership in a very literal way, through space cooperation.”¹²⁸

– U.S. secretary of state Antony Blinken, January 13, 2023

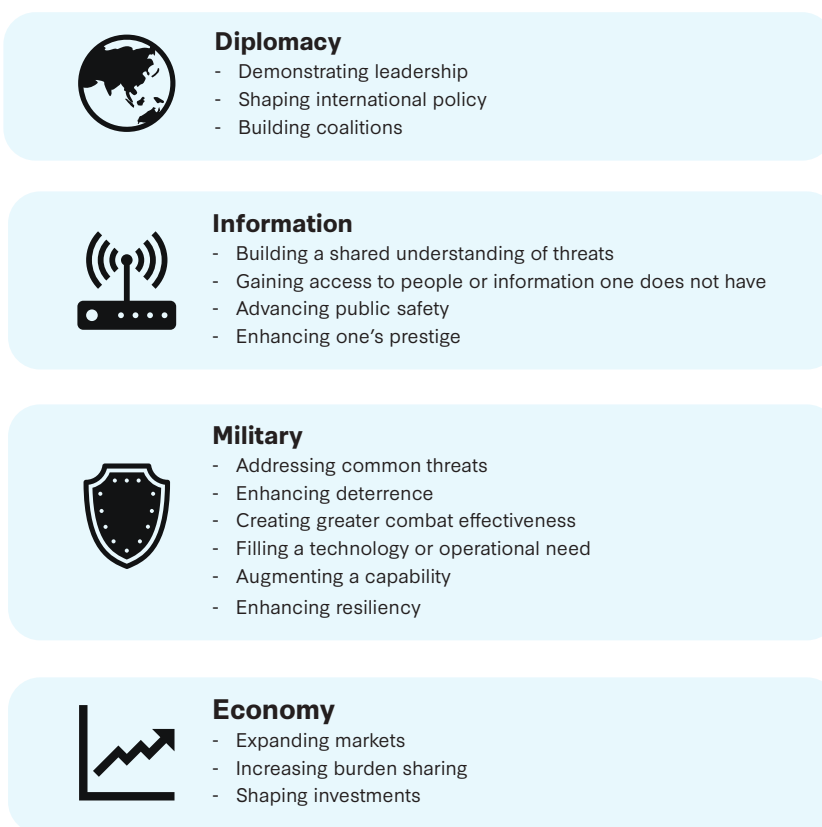
Space has become more prominent in high-level discussions between the United States and Japan, with political and military leaders from both nations committing to deepen cooperation in space security. This builds upon a long history of space cooperation, especially across civil space organization missions and organizations, which span human spaceflight, scientific research, satellite navigation and communication, and defense cooperation.¹²⁹

In January 2023, U.S. president Joe Biden and Japanese prime minister Kishida Fumio announced a bilateral space framework agreement, further cementing cooperation on space exploration, while also discussing the need to better align force postures in the space domain.¹³⁰ The U.S. and Japanese defense and foreign ministers now regularly highlight the growing importance of space to stability, security, and economic prosperity. In 2023, they committed to deepen cooperation “on space capabilities to strengthen mission assurance, interoperability, and operational cooperation.”¹³¹ Then, in April 2024, President Biden and Prime Minister Kishida went further, announcing a bilateral collaborative effort on a satellite constellation to detect and track missiles, including hypersonic glide vehicles.¹³²

Policymakers from both countries emphasize the increasing importance of space in their defense strategies, with space cooperation providing “an asymmetric advantage that our competitors can never hope to match.”¹³³ Japan plans to “radically expand the use of space systems for national security” while also playing a “proactive role in shaping norms for responsible behavior in outer space.”¹³⁴ The United States is prioritizing resilient space architectures, promoting responsible behaviors in the space domain, and expanding space cooperation to enhance information sharing and develop combined operations in space.¹³⁵

Figure 5: Cooperation: When National Interests Align

Bilateral and multilateral cooperation occurs when nations’ mutual interests align. Using the diplomatic, informational, military, and economic framework that classifies instruments of national power, mutual interests can fall into the following categories, as exemplars:



Source: Authors’ analysis.

Artemis Program and Artemis Accords

Japan is a partner of NASA’s Artemis program, which aims to explore the Moon and Mars. As part of the Artemis missions, the Lunar Gateway will serve as an outpost that will orbit around the Moon to stage deep-space exploration.¹³⁶ In November 2022, NASA and the government of Japan announced specific contributions under the “Gateway Implementing Arrangement.” Under this arrangement, Japan agreed to contribute critical components to the Lunar Gateway’s

International Habitation module and the JAXA HTV-XG spacecraft to launch and deliver resupply missions for the Lunar Gateway. Japan's contributions to the International Habitation module will include an Environmental Control and Life Support System, cameras, functions for thermal control, and batteries for additional modules. In return, NASA agreed to fly a JAXA astronaut as a Lunar Gateway crew member on NASA's Space Launch System rocket.¹³⁷ The Japanese Ministry of Education, Culture, Sports, Science, and Technology later added that the country seeks to have a Japanese astronaut land on the Moon in the late 2020s as the first non-American on the Moon, an aim that was reinforced by Prime Minister Kishida and President Biden in a 2024 state visit where Japan's anticipated role of providing the first international Artemis astronaut on the lunar surface was formalized.¹³⁸

Separately, Japan was one of seven original signatories to NASA's Artemis Accords in 2020. The accords, which are separate from the Artemis campaign, are a set of "principles for cooperation in the civil exploration and use of the Moon, Mars, comets, and asteroids for peaceful purposes."¹³⁹

2023 Space Collaboration Agreement

In January 2023, the U.S. secretary of state and Japan's minister of foreign affairs signed a space collaboration agreement titled "Framework Agreement Between the Government of Japan and the Government of the United States of America for Cooperation in Space Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, for Peaceful Purposes."¹⁴⁰ The specific areas of collaboration emphasized include space science (including Earth and lunar science),

space exploration, technology, and mission assurance and safety.¹⁴¹



Secretary of State Antony J. Blinken and Secretary of Defense Lloyd J. Austin III greet Japanese foreign minister Hayashi Yoshimasa and Japanese defense minister Hamada Yasukazu prior to the 2023 U.S.-Japan Security Consultative Committee at the Department of State.

Photo: Staff Sgt. Jackie Sanders/Office of the Secretary of Defense Public Affairs/DVIDS

U.S.-Japan Security Consultative Committee

The U.S.-Japan Security Consultative Committee (SCC), often referred to as the "2+2" meetings and comprising both countries' foreign and defense ministers, is a long-standing bilateral forum for discussing security and defense cooperation. The joint statement from the January 2023 SCC meeting emphasized alliance modernization and renewed both sides' "commitment to deepening cooperation on space capabilities to strengthen mission assurance, interoperability, and operational cooperation." It also, notably, included a declaration that "attacks to, from, or within space . . . in certain circumstances, could lead to the invocation

of Article V of the Japan-U.S. Security Treaty,” which commits the United States to defend Japan if attacked by a third party.¹⁴²

U.S.-Japan Comprehensive Dialogue on Space

To cement both nations’ commitment to bilateral space cooperation, Japan and the United States engage in a Comprehensive Dialogue on Space. This is one of only two comprehensive dialogues that the United States maintains, and it is the most longstanding one. In March 2023, the eighth meeting was held in Tokyo, which included representatives from the governments of both nations and, for the first time, industry participants.¹⁴³ Policymakers discussed national security, including Japan’s 2022 NSS, as well as commercial and civil space cooperation, with an emphasis on space situational awareness.¹⁴⁴ The ninth dialogue is planned for the summer of 2024 in Washington, D.C.

U.S.-Japan Military Cooperation

“We cannot secure stable use of space by unilateral effort, therefore, it is essential to cooperate with the U.S. and our partners at all stages from peaceful time to armed contingencies.”¹⁴⁵

– Major General Yajima Masahito, director general of the Defense Planning and Policy Department, Air Staff Office, Japan Air Self-Defense Force

Several recent organizational developments have marked closer space ties between the two countries, especially at the operational level. The first instance of operational military cooperation stems from a 2020 agreement between the U.S. Space Force and Japan’s Office of National Space Policy, whereby two U.S. payloads will be hosted on Japan’s QZSS satellites. The U.S. payloads have space domain awareness optical sensors and have been delivered to Japan to launch on the QZS-6 and QZS-7 satellites from Japan.¹⁴⁶

In December 2023, Japan participated in its first senior-level meeting of the Combined Space Operations initiative, where it joined a cohort of 10 countries working to improve multilateral space coordination, information sharing, and operational cooperation for the space domain.¹⁴⁷ Political and military leaders from each of the 10 nations’ departments of defense meet annually to deter conflict, optimize resources, facilitate information sharing, and enhance mission assurance and resilience.¹⁴⁸ The anticipated establishment of a U.S. Space Force command element in Japan in 2024 would facilitate further opportunities for bilateral training, exercises, and operational collaboration. This would be similar to U.S. Space Force command elements established in the Republic of Korea under the Space Force component supporting U.S. Indo-Pacific Command and in Tampa, Florida supporting U.S. Central Command.¹⁴⁹

These developments supplement an already established liaison officer initiative whereby Japan places a space liaison officer with the U.S. Space Command (USSPACECOM).¹⁵⁰ The U.S. Space Force (USSF) has an officer assigned to U.S. Forces Japan; that role is expected to transition into a new USSF field command element in Japan later in 2024.¹⁵¹ As combined training and operational collaboration increases, further embedding liaison personnel would likely be welcome

by Tokyo, including the placement of USSF or USSPACECOM personnel with the Japan Space Operations Group.

To further integrate military approaches, Japan and the United States participate in hosted wargames. Each year since 2001, the United States has hosted an annual Schriever Wargame, originally organized by Air Force Space Command and now led by the U.S. Space Force. Japan has been a participant since 2018. Participation in these wargames builds confidence and insight into the strategy, capabilities, and decisionmaking of other nations, especially with regard to space threats. In 2024, Japan joined NATO members and other partners to participate in the French AsterX exercise.¹⁵²

Deepening Cooperation in Space Security

There are abundant opportunities for deepening U.S.-Japan cooperation and collaboration in space security, building on the decades-long relationship in civil and commercial space cooperation and the nascent security activities underway.

Alignment on threats and strategic interests, guidance from senior leadership and political support, and the commitment of personnel and resources create a window to advance a comprehensive agenda for space security cooperation. Questions of implementation remain, including how to translate high-level policy intent into meaningful actions to deliver space capabilities, create interoperability across systems, and enhance mission resilience. This must be done while leveraging the complementary strengths that both countries bring forth, building greater collective knowledge and expertise in the space domain, strengthening the industrial base and promoting a culture of innovation, and expanding the talent pool across each country. But collaboration should not be pursued just for collaborations' sake—there must be intent behind actions to ensure that the respective industrial, economic, and security needs are met for both nations.

This section describes several areas where the combined, complementary strengths of both nations' public and private sectors can be brought together to deliver concrete space capabilities to advance both U.S. and Japanese national and collective security interests. These areas span collaborative efforts at the policy, operational, and technical levels and involve the governments, industry, and academia of both countries. The following section unpacks persistent challenges to greater cooperation. Each area includes discussion of the authors' observations and findings from their research, interviews, and workshops, as well as the recommendations that follow therein.

Collaboration should not be pursued just for collaborations' sake—there must be intent behind actions to ensure that the respective industrial, economic, and security needs are met for both nations.

Cultivating Human Development

Japan's past limitation that space be used only for nonmilitary purposes led its space ecosystem to grow around JAXA, civilian space missions, and civilian-led space science and technology research. This has been reinforced in the bilateral relationship through long-standing cooperation across space science and human spaceflight, as well as through expansion to cislunar and the Moon through NASA's Artemis program and the Artemis Accords. Yet while Japan grew into one of the world's most advanced spacefaring nations, its defense institutions and supporting industry have little exposure to, and limited experience in, military space missions.

This includes establishing requirements for military space systems, developing and utilizing those systems, and integrating them into joint military operations. Building such knowledge and proficiency starts with education and training, which serves as the foundation for the space security relationship. Japanese officials describe this as “human development,” expressing a near-unanimous view across government, industry, and academia on the need for space education and training for both military and civilian personnel.¹⁵³

To jump-start this human development, U.S. space institutions across government and academia could partner with Japan's MOD and the JASDF to provide civilian and military personnel with greater space domain knowledge. This could include invitations by the U.S. Space Force to attend “Space 100/200” courses and Defense Acquisition University space acquisition courses. U.S. academic institutions, partnered with Japanese academia, can also serve an educational role. The USSF has partnered with Johns Hopkins University's School of Advanced International Studies for its professional military education, which—in addition to other academic institutions—has established executive education courses in space security. To maximize these education and training opportunities, leaders on both sides can also encourage greater foreign language proficiency across their ranks.

Several Japanese experts further describe a shortfall in “human resources”—the workforce required to support Japan's space security vision—and cite the need for more engineers (e.g., aerospace, mechanical, electrical, and software engineers).¹⁵⁴ In the near term, such a human resources shortfall is expected to place greater strain on JAXA, given its current workforce level and resources and the industrial base that supports it. Not only is JAXA supporting an expanded space exploration program, including the development of the Artemis Lunar Gateway and lunar rover, it is being asked to partner with the MOD to support the Space Security Initiative and develop many of the satellite systems described in it.¹⁵⁵ This includes delivering the first domestically designed and produced SDA satellite for the JASDF.¹⁵⁶



GITAI is a start-up company developing robots that can perform general tasks and experimentations in space.

Photo: Takashi Aoyama/Getty Images

To help educate and inform MOD space professionals, JAXA assigns subject matter expert employees to the MOD defense policy bureau. However, it is important for the MOD to retain its own cadre of space professionals who are experts in space security policy, operations, and technology, as JAXA supplying personnel to the MOD is not a long-term solution to the MOD's human resource shortfalls. Given JAXA and its predecessor organizations' long-time focus on nonmilitary activities, it also lacks knowledge and experience with military requirements and operations, which differ in significant ways from those in a civil space context.

There is a call to “expand and enhance” JAXA's human resources and to recruit, train, and

retain space professionals who can develop, build, and operate both civil and national security space missions.¹⁵⁷ Over the next 10 years, the Japanese government has committed ¥1 trillion toward developing Japan's domestic space industry. This Space Strategy Fund is aimed at supporting JAXA and the expansion of Japan's space industrial base.¹⁵⁸ Japan's expanding start-up ecosystem should be nurtured as a pathway to bring greater technical talent into the space ecosystem. As one 2023 study notes, “Japan's start-up ecosystem is currently able to draw top-tier talent to high-growth start-ups at a scale that was not possible fifteen years ago.”¹⁵⁹ However, expertise takes time to cultivate. Partnerships with U.S. space and technology companies, especially on cooperative projects involving codevelopment or coproduction, can hasten this knowledge and experience while also adding needed capacity.

Also observed within Japan was the lack of space-related media and think tanks in contrast to the robust presence of media in the United States dedicated to reporting on civil, commercial, and military space developments. Public discussion about space programs and plans, technologies and start-ups, and the security and economic benefits of space may also lead to excitement, understanding, and greater support among the Japanese population. Without this public reporting, space would not receive broader attention and inclusion in conversation. Both U.S. and Japanese government and industry leaders could be more deliberate in their engagement with traditional Japanese media on space initiatives and collaborative efforts to build understanding and broader encouragement.

Exercising and Training the Space Force and Joint Force

From early warning and intelligence to border and maritime security, humanitarian assistance, disaster relief operations, and high-end combat, space capabilities contribute to nearly every

aspect of modern military operations. As Japan develops the space capabilities outlined in the SSI, it will be vitally important not just for the JASDF to build space proficiency, but for the broader Japanese joint force to understand how space capabilities enable its missions. In turn, such joint force learning will shape requirements on what the SDF needs from space systems moving forward. This learning is built through education, as discussed above, and integrating space elements into exercises and training.

In these exercises is where theory meets reality. This is where both Japanese and U.S. forces work through the space capabilities each side can bring, the quality and timeliness of information needed from those systems to make decisions, and how that information is presented and delivered to users. These exercises can also be used to build interoperability between systems and identify any factors that may limit mission effectiveness.

Space capabilities can be increasingly integrated into bilateral and multilateral joint exercises, such as Keen Edge, Keen Sword, and Rim of the Pacific, and into routine service-specific and joint training, at both the command and field levels. Notably, Keen Edge 2024—a command-level bilateral exercise between U.S. forces and the Japanese Self-Defense Forces to enhance joint operations and interoperability for crises and contingencies in the Indo-Pacific—featured USSPACECOM participation for the first time.¹⁶⁰ Military commanders will find that space capabilities play a key yet less visible role in joint force employment and can be a useful contributor as Japan evolves its command and control structures and stands up a joint operations command.



Guardians from the U.S. Space Force's 16th Electromagnetic Warfare Combat Detachment and key leaders from the Japan Air Self-Defense Force's Space Operations Group meet during the first Bilateral Space Professional Enhancement Collaboration at Camp Kinzer, Okinawa, December 6, 2022.

Photo: Judi Tomich/U.S. Space Force

Within the military space community, JASDF officials noted the value of taking part in the U.S. Space Force Schriever space wargame series and USSPACECOM's Global Sentinel exercise, both of which expose participants to multilateral space operations scenarios. Such wargames and realistic scenarios, along with immersive tabletop exercises and collaborative studies, were cited as ways to improve Japanese understanding of space threats and the principles of operating in space for those new to the domain. They can also be leveraged by space policymakers and operators to explore scenarios that might lead to “the invocation of Article V of the Japan-U.S. Security Treaty,” as discussed at the 2023 U.S.-Japan Security Consultative Committee meeting, including considerations regarding the decisionmaking processes and information needs of both countries.¹⁶¹

Furthermore, JASDF officials recognize that their space forces will need to gain greater operational experience as they field satellite constellations that support Japan's national security. As the MOD and JASDF expand into new mission areas in which satellites in orbit can maneuver to characterize other objects in space, such as space domain awareness, additional training will accelerate their ability to effectively employ the systems once they are fielded. Japan's first SDA satellite is planned for launch in FY 2026, so now is the time to commence such training.¹⁶²

Such training and sharing of operational best practices support a policy goal shared by both countries to advance responsible behaviors in space, including conducting activities in space that promote the safety and sustainability of the domain. These capabilities also benefit the United States by having additional assets capable of identifying unfriendly or unusual satellite behaviors in orbit.

Private sector Japanese and U.S. satellite operators have also gained greater operational insights from their participation in space wargames and exercises. For example, Japanese industry participated in a space situational awareness tabletop exercise through USSPACECOM's Joint Task Force-Space Defense Sprint Advanced Concept Training initiative, which has since been renamed the Joint Commercial Operations (JCO) cell.¹⁶³ The satellite operators were presented with operational problems such as how to respond to others' satellite close approaches and maneuvers, satellite breakup events, etc. These scenarios challenged their thinking, helped them refine their own best practices, and forged key relationships with U.S. government and industry partners. Exercises such as these are all the more critical given Japan's participation in USSPACECOM's JCO cell. As industry becomes a trusted partner in fielding and operating satellite systems, its involvement in joint exercises will continue to become more important.

SHARING DATA AND INFORMATION

Information sharing helps build a more complete picture of emerging threats and enables each country to act with more fulsome knowledge than it would be able to individually. Whether for missile defense, counterstrike, space operations, or cyber defense, information will need to be disseminated to users in a timely manner and increasingly at machine speeds.

An information exchange model often cited is the trilateral mechanism between the United States, Japan, and South Korea that enables real-time data sharing on North Korean missile launches.¹⁶⁴ However, in the space domain, both U.S. and Japanese officials shared observations that the current information-sharing paradigm is "largely emails and PDFs," not space threat information or operational data that both sides can access, analyze, and use. As both countries emphasize speed and adopt greater automation, advanced processing, and artificial intelligence and machine learning technologies, they must be able to share datasets on operationally relevant timelines, whether in terms of real-time data on the launch and trajectory of a threatening hypersonic glide vehicle or regarding the maneuvering and close approach of an unfriendly satellite.

Sharing space and cyber threat assessments is a logical start, especially with the increasing availability of both classified and unclassified threat assessments from the U.S. government, industry, and academia. Further, SSA and SDA data are strong candidates for enhanced sharing given the emphasis both countries place on these capabilities. The Japanese MOD, the JASDF, and

JAXA maintain terrestrial SSA sensors and are developing an SDA satellite. The USSF is expanding its global space surveillance network (working with allies) and SDA satellites, and U.S. commercial SSA providers add capacity with shareable data and orbital analysis. The two nations are collaborating further on a USSF SDA payload to be hosted on a Japanese QZSS satellite, yet no concrete plans have been made to provide that SDA data to Japan.

This is a clear area in which both parties bring data and information to the table that can increase collective capacity to identify unusual and unfriendly behaviors and attribute threats in the space environment. Such information sharing can also support both nations' leadership efforts to reduce space threats through the promotion of principles and norms for responsible space activities. Further, because SSA is an essential capability for a majority of spacefaring nations, establishing a cooperative information-sharing mechanism can provide the building blocks for greater multilateral information sharing on space activities going forward, including between Japan, South Korea, and the United States.

SSA and SDA operations and operational data remain highly classified within the U.S. Department of Defense (DOD). It will be critical for U.S. policy officials, working with USSPACECOM, to update and streamline policy permissions for sharing such information and to accelerate the implementation of the DOD's revised classification policy for space programs and operations.¹⁶⁵

COMPLEMENTARY INDUSTRIAL BASE STRENGTHS

Both nations emphasize the importance of partnerships between U.S. and Japanese industry for delivering military space capabilities within relevant timeframes. However, there will be a natural tension between the countries as each will seek to expand its own industrial base and decide what it is willing to rely on the other for. Moreover, while Japanese industry is well versed in building exquisite scientific satellites for JAXA and contributing to civil missions, the contrasting scale and timeline shift for developing, producing, and fielding military systems is substantial.

It is useful to consider the unique strengths maintained in each nation's industrial base that lend themselves to collaborative efforts.

In areas where timely action is particularly critical, the quickest pathway to a near-term, interoperable capability may be for Japan to acquire space systems from U.S. aerospace and defense companies that maintain an institutional and operational knowledge base that may be impossible to scale from scratch. But a strict "buy American" approach does not meet Tokyo's policy objectives of security and industrialization that would help it strengthen its industrial base and attract talent. Likewise, a Japanese domestic-only approach in which U.S. industry does not see benefit or expanded market access is equally unproductive, especially as Japan seeks to speed up its technical and operational proficiency in developing and utilizing military space systems. Therefore, it is useful to consider the unique strengths maintained in each nation's industrial base that lend

themselves to collaborative efforts that enhance the alliance's overall deterrence and defense posture in addition to meeting industrial development needs.

A strength across U.S. industry is its institutional knowledge of military space systems, particularly from its aerospace and defense primes, which have been designing, building, and operating military satellite systems for decades. U.S. industry brings experience in systems integration of large satellite constellations and in constructing the ground infrastructure necessary for data processing, analysis, and connectivity with users on militarily relevant timelines. In some areas, such as LEO intelligence, surveillance and reconnaissance, and communications satellites, U.S. industry has scaled up manufacturing capacity with hot production lines. Sharing lessons learned from past projects and providing systems integration and ground architecture expertise may prove invaluable as Japanese industry takes on large national security space constellations with pressure to deliver in a short time span.

Japanese industry holds a particular strength in satellite components, with proven hardware in areas such as transponders, amplifiers, optical links, and solar arrays. These components, built by Japanese companies such as MELCO and NEC, are used in hundreds of satellites from dozens of nations around the world. As the United States seeks to increase supply chain resiliency, it should look to these Japanese component providers as part of its broader trusted supplier base. Further, Japan has long maintained strength in certain cutting-edge technology areas, such as robotics and advanced manufacturing, especially in the automotive sector.¹⁶⁶ There are best practices to be drawn from these sectors and applied to both Japanese and U.S. satellite production.

On the commercial services front, both nations have long-standing commercial SATCOM operators, such as Sky Perfect JSAT and Viasat, that are being joined by a new crop of commercial space companies. Bolstered by private capital, these companies are providing data, analytic products, and services in areas like Earth observation, space situational awareness, SATCOM, and in-space servicing. They offer capabilities that can be integrated into each country's space architectures, providing shareable data and additional capacity.

According to a 2023 report on trends in start-up space investments by U.S. analytics firm BryceTech, roughly 50 percent of all investors in start-up space companies since 2000 were from the United States (43 percent) and Japan (7 percent).¹⁶⁷ BryceTech further noted that “[p]articipation from international investors has been increasing since 2018, primarily in China, Japan, and the United Kingdom.”¹⁶⁸ Such capital, including reciprocal investments in each nations' space sectors, can boost the development of the commercial space capabilities and services described above to achieve broader alliance objectives vis-à-vis space.

Similarly, Japan's Space Strategy Fund has the potential to strengthen Japan's broader space and technology ecosystem so that it can more substantially contribute to the space security needs of both countries. Ideally, the fund would focus not only on big ticket hardware like space launch vehicles, but also on stimulating Japanese start-ups, advancing dual-use technologies, and developing space data analytic tools and services that can support the MOD's objectives, as well as encouraging Japanese firms that are beneficiaries of the fund to partner with U.S. firms.

While U.S. and Japanese start-ups are one source of innovation and disruption within the space ecosystem, government agencies are also introducing disruptive acquisition approaches and business models to traditional space acquisition programs. As the developer, acquirer, and operator of the next-generation Proliferated Warfighter Space Architecture (PWSA), the U.S. Space Development Agency—along with its industry partners—is well equipped to share insights on acquisition best practices, the development of open source and modular components, and spiral development methods to incrementally deliver new capabilities over time.¹⁶⁹

Finally, while it may be expedient to focus on partnering constructs within the space ecosystem, what each country needs most from the other may not be space specific. For example, while Japan may be best served by U.S. industry aiding in the system integration of its military satellite constellations, the United States may be best served leveraging Japanese industry for overseas ship repair and maintenance, or mass production of drones and munitions, given Japan's strengths in advanced robotics and manufacturing.¹⁷⁰ Expanded thinking on industrial base cooperation and broader “win-win” approaches that leverage each nation's complementary strengths should be encouraged.

Architecting National Security Space Constellations

With its Space Security Initiative, Japan for the first time presents a road map for a multifaceted space security architecture with constellations of satellites performing missile detection and tracking, information collection, communications, positioning and navigation, and space domain awareness, supported by a suite of space launch vehicles. Meanwhile, the United States is in the midst of major changes to its national security space architecture, prioritizing mission assurance through diverse, resilient space architectures that includes greater investment in proliferated missile warning and missile tracking (MW/MT); intelligence, surveillance, and reconnaissance; communications constellations of smaller, less exquisite satellites; and space protection and space control capabilities.¹⁷¹ Both countries emphasize that government systems and commercial services, including new areas of space commercialization, will be needed to implement these strategies.

Washington and Tokyo appreciate that their alliance and partnerships provide them with an asymmetric advantage and that their cooperation “broaden[s] the number of systems collectively available for space operations, both on orbit and on the ground; strengthen[s] resilience; expand[s] our options for diplomatic and military responses; and complicate[s] an adversary's decision making.”¹⁷²

Both approaches are ambitious, resource-intensive endeavors. However, despite increased funding for space in both countries, delivering on these plans will be challenged over the long term by resource availability, industrial capacity, and labor accessibility in highly technical fields. Japanese experts emphasize that they need U.S. partners, as Japanese industry lacks sufficient human resources and would be hard-pressed to deliver space capabilities at the scale and timelines needed. While the United States brings deep experience in military space systems development and operations, it also seeks additional capacity for resiliency and filling gaps and new ways to share the burden. Moreover, as the U.S. approach transitions to proliferated architectures, there will be opportunities to create new concepts of operations, different technology development and production arrangements, and new partnering constructs.

The authors considered numerous ideas on architecture cooperation across different space mission areas, informed by discussions, briefings, and workshops with U.S. and Japanese government and industry representatives. Several of those will be discussed below.

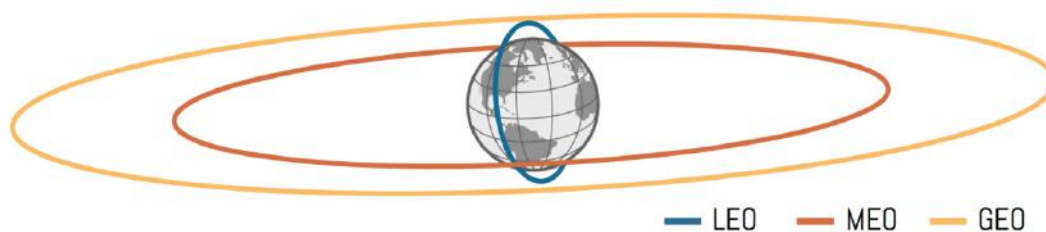
While all areas described in the SSI could be candidates for greater bilateral cooperation, missile detection and missile tracking, tactical surveillance, reconnaissance and targeting capabilities, protected SATCOM, and SDA stand out because of the urgency to have sufficient defensive capabilities deployed by FY 2027 described in Japan's strategy documents.

MISSILE DETECTION AND TRACKING

The expanding missile arsenals of North Korea and China, including advanced ballistic and cruise missiles and HGVs, have made space-based missile detection and tracking one of the more pressing areas of need for the United States and Japan. This led leaders from both countries to commit, in April 2024, to pursue “bilateral collaboration on a Low Earth Orbit detection and tracking constellation for missiles such as hypersonic glide vehicles, including potential collaboration with U.S. industry.”¹⁷³

For the United States, a shift in approach is underway toward a proliferated MW/MT architecture with satellites in multiple orbits to increase resilience against growing space threats that also takes advantage of commercially derived components and practices. The U.S. Space Force, through its Space Development Agency, is building out hundreds of satellites in LEO through the PWSA; the USSF and MDA are building medium Earth orbit (MEO) satellites, and the USSF is replacing its GEO and high Earth orbit space-based infrared systems through a next-generation overhead persistent infrared (OPIR) program.

Figure 6: Popular Orbit Regimes



Source: Aerospace Security Project.

Japan aims to develop a sovereign capability and, while there is interest in LEO collaboration, it has not yet coalesced on an overall architectural approach. To date, Tokyo has relied on missile warning data shared from U.S. space-based infrared systems and defense support program missile warning satellites in GEO and high Earth orbit and its own ground-based radars. Some in Japan question the need for a global architecture and instead argue that regional needs can be met with GEO systems that provide persistence and sensors that can monitor the region. Japanese industry has proficiency in building large satellite buses that could be integrated with an OPIR payload for GEO operations. Others argue that LEO or MEO add complementary capacity and increase resiliency,

with opportunities to build numerous smaller satellites and supply satellite components over time with more regular technology refreshes. Regardless of the approach, such an architecture is a significant, long-term resource investment. At this point, it is important not to foreclose options, but policymakers, program officers, and political leaders must make decisions on requirements in short order.

With this backdrop, the authors offer a few ways to think about cooperation going forward. Building and operating a space-based missile detection and tracking architecture is new for the MOD and SDF, their acquisition organizations, and the industrial base. Phased approaches that enable Japan to better learn the military mission and new target set (e.g., HGVs) to inform its overall requirements and to build technical and operational proficiency over time came up repeatedly in workshops and discussions led by the authors. It was also argued that, of all space mission areas for U.S. industry to heavily engage in, missile detection and tracking is particularly symbiotic. It is one of the few technology areas that Japan's industrial base has not yet developed, and it cannot be acquired through commercial services. Further, it is an area in which the United States is planning to field hundreds of satellites in a global constellation and is a natural extension of the already close missile defense relationship.¹⁷⁴

A partnership between the MOD and the U.S. Space Development Agency centered on the tracking and transport capabilities of the PWSA LEO architecture offers an attractive pathway for such phased collaboration and contributes to greater overall resiliency. With its planned architecture of smaller, less expensive satellites numbering in the hundreds, PWSA provides an incremental approach for Japan to learn the mission and to scale its small- to mid-sized satellite development and production (building “reps and sets”).

Described herein are several recommendations specific to missile detection and tracking cooperation:

- In the near term, a PWSA ground station could be deployed in Japan and PWSA data shared with Japan. Building off the current early warning data-sharing model, this would allow both Japanese and U.S. users to gain familiarity with PWSA tracking data; build experience processing, analyzing, and fusing it with other missile early warning data; and provide feedback on the data quality, latency, and connectivity needed to support decisionmaking and missile defense systems.
- In the near term, a joint technology and production demonstration program could be pursued for missile detection and tracking and transport layer satellites. This would give both industrial bases opportunities to explore different partnering arrangements while also building Japanese industrial capacity. For example, Japanese optical communications payloads could be integrated into transport satellites, and coproduction or licensed production arrangements could be explored for both transport and tracking layer satellites. Both constructs could fit within the Space Development Agency's “future capabilities initiative,” in which incremental improvements can be on-ramped into future tranches of PWSA capability. Japanese optical communications payloads should be interoperable by design with PWSA optical communications, which are open source.

- Such a joint technology development program could also include a coordinated test campaign and sharing of test data, especially involving performance against HGVs. Japan could participate in U.S. PWSA testing activities and, likewise, the United States could take part in Japan's testing scheme as it flies its own military infrared (IR) payload demonstration on a future HTV-X mission.¹⁷⁵
- Over the mid to long term, Japan could scale up its domestic industrial capacity and technical proficiency to build a sovereign operational constellation that is informed by a clearer understanding of mission needs and is complementary to and interoperable with the U.S. PWSA.

If timely fielding of an operational capability is the driving need, Japan could lean heavily on U.S. industry early on, taking advantage of active satellite production lines and minimizing time for design and development. As Japan builds experiential knowledge and domestic production capacity, this dependency would wane.

If Japan decides to pursue a GEO architecture, industry options such as a Japanese satellite bus paired with a Japanese or U.S. OPIR payload could be examined. For example, Japan has funded plans to fly a military IR payload demonstration on a future HTV-X launch that could be a candidate. While the payload is likely to be highly capable, U.S. experience shows that such new design and development efforts are complex, costly, and take time. While satellite interoperability between U.S. and Japanese systems will be harder with two different designs, opportunities for fusing data in ground stations from GEO and LEO systems could be explored.

It must be emphasized that partnerships beyond satellite constellations are important in this mission area. For missile detection and tracking—a mission that must operate on militarily relevant timelines—the ground architecture, data processing and analysis, and connectivity to missile defense systems can oftentimes be more challenging than satellite development and will demand equal attention. There is just as much opportunity to forge industry partnerships in ground station processing, advanced analytics, battle management, and command and control functions as there is in satellite development and production.

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INTELLIGENCE, SURVEILLANCE, RECONNAISSANCE, AND TARGETING

The SSI describes the need to “advance target detection and tracking capabilities” for Japan's standoff defense capability and maritime domain awareness.¹⁷⁶ This includes better understanding the surrounding terrestrial, maritime, and air operational environments, building an intelligence

picture of threats, and providing indications that the threat has changed. Further, Tokyo's standoff counterstrike systems will require real-time targeting information provided by intelligence, surveillance, and reconnaissance systems.

Various Japanese representatives shared that the government plans to invest in a constellation of dozens of electro-optical (EO) and synthetic aperture radar (SAR) satellites that would provide ISRT capabilities in day, night, and all-weather conditions.¹⁷⁷ These systems would also play a role in detecting and tracking missiles prior to launch, such as tracking adversary mobile missile movements in and out of garrison, mapping missile-related facilities, and building patterns of activity among missile units.

In the area of ISRT, Japan has a strong sovereign foundation on which to build. The Cabinet Satellite Intelligence Center, working with Japanese heavy industry, has built and operated EO and SAR satellites as part of its IGS constellation for over two decades, with plans to further expand the constellation. Japan's domestic commercial remote sensing industry is emerging, with both mainstays and start-ups building commercial imagery satellites, especially in SAR, and aiming to sell data products as a service.

Japanese industry and the Japanese government have a strong pedigree in this mission area. They also have a long history of working closely with U.S. intelligence agencies and U.S. space companies providing commercial imagery. Therefore, cooperation efforts should focus on adding capacity and resiliency to the alliance, engaging in ISRT data sharing, building greater analytic and multi-intelligence insights, and gearing the overall architecture toward supporting military requirements and delivering actionable information on operationally relevant timelines.

The authors identified key partnership areas across ISRT, which are listed below:

- Both countries see benefits in the greater adoption of commercial satellite data and analytic services to support traditional military missions as well as to enhance maritime domain awareness, support humanitarian assistance and disaster relief operations, and track global trends and environmental changes. The MOD is likely to increase its FY 2025 budget for the acquisition of commercial imagery, and the U.S. Space Force has begun leveraging commercial data sources.¹⁷⁸ This is an important trend to continue, not just to support national security priorities and decisionmaking but also to strengthen the global competitiveness of both countries' commercial space sectors, which are facing increasing competition from Chinese competitors.¹⁷⁹
- Although Japanese industry has experience building EO and SAR satellites, a demand for scale and the timely fielding of systems will necessitate partnerships that abide by Tokyo's "security and industrialization" policy while also leveraging U.S. industrial scale and systems integration expertise. Such partnering arrangements could include coproduction and license production agreements between U.S. and Japanese space companies and the acquisition of commercial EO and SAR data or dedicated access to commercial satellite constellations. In the longer term, Japan could scale up its domestic industrial capacity to sustain this constellation on its own. In

order to better collaborate, U.S. firms—and the U.S. government—would benefit from a greater understanding about Japan’s general architecture envisioned for an ISRT constellation.

- ISRT is another mission area in which collaboration will need to be fostered beyond satellites, with a commensurate focus on ground systems, data processing, and analysis, as well as interoperability across those ground systems. This includes developing a ground architecture and processes for tasking, collecting, processing, exploiting, and disseminating data from the EO/SAR constellation. Further, the volume of data from a large ISRT constellation will be immense; there will not be enough human analysts in Japan or the United States to scrutinize each image for military intelligence value. Partnerships that emphasize data processing, automation, and the development of advanced analytic tools, including the use of artificial intelligence and machine learning, can be encouraged.
- To enable one of its priority mission areas, counterstrike, the JASDF recognizes that it will need to improve proficiency in military intelligence analysis and target development, such as the targeting of “vessels and landing forces.”¹⁸⁰ The Bilateral Intelligence Analysis Cell, established in November 2022, aims to do just that, with Japanese and U.S. analysts at Yokota Air Base jointly analyzing video feeds from remotely piloted aircraft. It could be expanded to incorporate space-based remote sensing data.

The authors were encouraged to explore ways that data sharing between CSICE and the MOD and SDF could be increased, as Japan’s defense and intelligence institutions remain highly stovepiped. The authors note that this remains a perennial challenge within the United States as well—how to ensure that the intelligence community supports both national decisionmakers and warfighters. The private sector can be a pathway for breaking down stovepipes. One additional area of collaboration across U.S. and Japanese companies could be demonstrating the fusion of multiple sources of information—not just EO and SAR, but also radiofrequency signals collection, hyperspectral imagery, and open-source intelligence. Industry, and particularly start-ups, are not beholden to the same stovepipes that governments are. They may be able to make greater progress on demonstrating the higher-value intelligence insights gained from multi-intelligence concepts of operation.

SPACE DOMAIN AWARENESS AND COUNTERSPACE

Both countries have advanced capabilities in tracking and monitoring space objects and debris. They are increasing their SDA investments as threats to space capabilities grow and the domain becomes more congested. SDA is already an area of active partnerships across both governments and industries. As described earlier, a 2020 agreement between the U.S. Space Force and Japan’s Office of National Space Policy led to two U.S. SDA payloads being hosted on Japan’s QZSS satellites (which have not yet launched). Industry partnerships are also being cemented, such as the agreement Northrop Grumman and IHI Aerospace struck in March 2023 to build small maneuverable satellites and other capabilities to enhance SDA.

An earlier section of this analysis described opportunities to enhance data sharing to improve the accuracy and coverage of space object tracking, reducing the risk of collisions and enhancing overall space traffic management. Described here are areas of cooperation centered



U.S. Space Force sends two SDA payloads to Japan.

Photo: U.S. Indo-Pacific Command

on SDA capabilities and infrastructure and joint initiatives focused on debris mitigation and removal.

- Unlike the United States, Japan is not bound to decades-old space object tracking systems and catalog systems. Japan has an opportunity to take a clean-sheet approach to building a common operating picture of the space domain by leapfrogging ahead: leveraging state-of-the-art commercial technology and fusing space surveillance data from both the Japanese and U.S. government and commercial sources, including data from JAXA ground optical telescopes and radars and commercial data (e.g., LeoLabs, Slingshot,

and COMSPOC). Further, such a capability could support government and commercial satellite operators alike, providing detections of close approaches by other satellites, building patterns of activity by unusual or unfriendly objects in space, and analyzing the risk of satellite collisions.

- Other hosted payload concepts in addition to the USSF SDA-QZSS partnership could be explored. Beyond the IHI-Northrop Grumman partnership to develop a joint SDA capability, other Japanese and U.S. companies are studying concepts that would integrate SDA payloads onto next-generation SATCOM systems and provide them to governments as a service. The recent announcement that the QZSS constellation will expand to 11 satellites provides an opportunity for further hosted SDA payloads.¹⁸¹
- Both countries could also consider coordinated operations between their SDA satellites—including Japan’s SDA satellites, the first of which is planned for launch in 2026—and U.S. Geosynchronous Space Situational Awareness Program (GSSAP) satellites. Such coordinated operations would aid in training, add capacity, and reduce the duplication of target coverage. Japanese scholars raised the idea of SDA satellites and U.S. GSSAP satellites “operating in coordination,” particularly once Japan’s second SDA satellite is fielded, as it is anticipated to have increased “mobility and imaging capability.”¹⁸²
- Japan is a leader in developing technologies for debris removal, including active debris removal with technologies such as laser ablation, while the United States has extensive experience in space operations. Joint demonstrations and missions could validate active debris removal technologies while also advancing new operational concepts for debris mitigation.

Both governments should also explore options to cooperate on space protection and build a baseline understanding of how each side views counterspace capabilities, although the topic and any capabilities will be highly sensitive. The JSOG possesses equipment to detect satellite jamming that will be useful for government and commercial operators alike as electronic attack

weapons become more proliferated. Japan's NDS also lays the groundwork for the development of capabilities "necessary to disrupt or neutralize the opponent's use of space."¹⁸³ For example, as part of efforts to secure its use of the electromagnetic spectrum, Tokyo allocated funds in its FY 2024 budget for communication and radar jamming capabilities.¹⁸⁴ While further counterspace activities are likely to be highly classified by each country—and have long been treated as such in the United States—a general discussion on basic concepts, capabilities, and scenarios would be helpful to understand how each side thinks about the mission area and, importantly, to avoid miscalculations. The recent U.S. DOD push to lower classification barriers for certain space capabilities, along with Japan's membership in Combined Space Operations, provides an opening to overcome foreign disclosure barriers to sharing information with Japan.¹⁸⁵

PROTECTED SATCOM

Resilient satellite communications are also an area of emphasis in both countries. The United States has long operated a range of military SATCOM systems, complemented by commercial GEO, and is increasingly leveraging LEO broadband networks, with Starlink being the most prominent. Japan also has a long pedigree in satellite communications, leveraging Sky Perfect JSAT to develop and operate a network of GEO satellites for military use. Japan's SSI outlines plans for a multilayered and redundant SATCOM network across GEO and LEO orbits, with government and commercial systems meeting growing demands for connectivity across Japanese forces and weapons systems.

While SATCOM cooperation between Japan and the United States is already underway and Japan has established a line of effort dedicated to commercial LEO broadband, government officials have expressed particular interest in partnering on protected SATCOM that can provide secure anti-jamming and anti-spoofing connectivity, especially as adversaries increasingly employ electronic warfare systems and tactics. On the Japanese side, its Joint Staff C4 Systems Department (J6) is leading protected SATCOM investment efforts, while on the U.S. side, the Space Force's Space Systems Command (SSC) is leading the acquisition of protected tactical SATCOM capabilities across satellites, ground infrastructure, modems, and user terminals.

Partnerships in the area of protected SATCOM could be explored in the following areas:

- Notably, as part of the U.S. SSC protected tactical SATCOM program, SSC is funding a project to deliver protected tactical waveforms over commercial and international GEO and MEO commercial satellite constellations used by the U.S. military and coalition forces. This is an area in which Japanese SATCOM operators could be a partner to SSC in providing platforms for the maturation of this capability. Japan would then be in a prime position to scale this across its fleet of SATCOM assets used for security purposes.
- As Japan refines its acquisition approach for a protected SATCOM system, one model to consider is the approach recently taken by Australia to develop a sovereign military SATCOM capability. The government of Australia required bidding teams to comprise a prime contractor with extensive military SATCOM experience and several Australian businesses.¹⁸⁶ Such an approach could even lead to greater interoperability across U.S., Japanese, and Australian SATCOM systems.

- As Japan fields SATCOM capabilities with optical communications payloads, opportunities exist for cross linking between communications networks, especially if both sides can align on common optical interface standards. Japanese government and industry representatives expressed particular interest in connectivity between Japanese space assets and the PWSA transport layer, creating even greater resiliency and multipath options for data delivery to users.

Investing in Digital Transformation

Digital transformation is essential to maintaining a battlefield edge, creating greater mission effectiveness through data accessibility and connectivity across users and resilience and security in the face of threats to IT infrastructure and communications networks. The U.S. Department of Defense is implementing a cloud strategy and making investments in capabilities such as the Air Force and Space Force's Cloud One.¹⁸⁷ The U.S. Intelligence Community (IC) has long invested in a cloud-based architecture known as the IC Cloud that enables users to log into a desktop from any IC location and access mission-related information.¹⁸⁸

A similar digital transformation is underway in Japan, with senior leaders discussing how a government cloud network can enable information collaboration, crisis response, and increased operational efficiencies across the government.¹⁸⁹ Tokyo is attuned to the importance of digital resilience and continuity of operations, having experienced natural disasters affecting critical infrastructure such as the 2011 earthquake and tsunami that damaged the Fukushima nuclear power plant and seeing Russian cyberattacks against Ukrainian satellite and ground networks. With regard to the latter, Ukraine's successful migration of government and banking data to a cloud storage system in the initial days of Russia's invasion proved crucial to the continuity of Ukrainian government services.¹⁹⁰

Importantly, national security leaders in both countries recognize that the private sector leads in cloud storage, computing, and security, and they benefit by adopting these commercial services. Neither government can match the scale, speed, performance, and agility of commercial technology companies or their investment in digital infrastructure. These companies invest billions of dollars per year in cybersecurity alone, continuously improving their networks and cloud infrastructure while under constant cyberattack. Microsoft, for example, cites 1.5 million attempts a day to compromise its systems and invests roughly \$1 billion annually in cloud security.¹⁹¹

Neither government can match the scale, speed, performance, and agility of commercial technology companies or their investment in digital infrastructure.

The bilateral space agenda would be well served by including a digital transformation and cybersecurity component: an area that is not exclusive to space but would enable the sharing

of sensitive mission data and operational information. Japanese partnerships with U.S. digital infrastructure companies could be particularly valuable, with companies such as Amazon Web Services planning to invest ¥2.26 trillion (\$15.4 billion) to expand its Japanese cloud infrastructure by 2027 and Microsoft investing \$2.9 billion in Japan on AI and cloud infrastructure, workforce training, and a research lab.¹⁹²

Leveraging these investments, the MOD and the JSOG could identify their expanding space architecture, including space operations, command and control (C2), and space domain awareness data, as early candidates for migration to the cloud. This interest was conveyed to the authors in conversations with Japanese space operators who recognize the benefits of cloud for greater connectivity, data management, and network security. U.S. officials emphasized that information security is essential to expanding data sharing given the sensitivity of space technology, operations, and increasing cyber threats to satellite networks.

Compatible digital infrastructures with common interface standards would provide the technical foundation to facilitate greater connectivity and data exchanges between U.S. and Japanese networks. Working from similar digital infrastructures creates more direct opportunities to share cybersecurity best practices and also creates a more accessible and fulsome data environment for both countries to exploit using AI tools and other innovative applications.

The DOD is also modernizing the information-sharing networks it uses for international partners to securely access, process, and analyze sensitive mission data and to deliver insights to users. For example, U.S. Indo-Pacific Command (INDOPACOM) is accelerating the development of the INDOPACOM Mission Network, which is built off of the DOD's larger Mission Partner Environment program to share information and enable a common operating picture with allies and partners.¹⁹³ Another U.S. information sharing system, the Battlefield Information Collection and Exploitation System program, provides connectivity for sharing military intelligence information. Not only should the DOD be encouraged to leverage commercial solutions, taking advantage of the benefits described earlier, but also to expand access points within the MOD and the JASDF, including the JSOG.

Leading on Policy and in International Forums

Sustainability has long been a theme in Japanese society and policy. Tokyo is well positioned to continue extending this theme into space as a leading voice on enhancing the sustainability of space. Both Japan and the United States, through complementary and coordinated efforts in regional and international forums, can provide strong thought leadership and advance diplomatic initiatives on space sustainability, developing responsible behaviors in space and reducing threats to the peaceful uses of space.

Japan's actions bolster its standing in these areas. It has designated sustainability and resource management as clear priorities for space development. New space companies, such as Astroscale, have crafted business models around the removal of on-orbit debris and the disposal of spacecraft reaching their end of life. Others are developing remote sensing satellites to monitor climate change



Kamikawa Yoko, foreign minister of Japan, speaks to the press at a meeting of the Security Council at the UN Headquarters.

Photo: Lev Radin/Pacific Press/LightRocket via Getty Images

as well as the security environment. Tokyo is becoming a destination for important space symposia, including on the topic of space sustainability.

Diplomatically, Japan is increasingly exercising its voice on space security and sustainability issues. In April 2024, it joined with the United States to introduce a United Nations Security Council Resolution on Preventing Nuclear Weapons in Space, calling on nations to uphold their obligations under the Outer Space Treaty.¹⁹⁴ This resolution was drafted in response to news that Russia may be developing a satellite with a nuclear device that would be indiscriminate in its destruction of satellites in orbit. In September 2022, Tokyo pledged not to conduct destructive,

direct-ascent ASAT missile testing, working with the United States and other nations to promote international discussions on responsible behaviors in space.¹⁹⁵

Japan should be seen as a regional leader on space security issues and as a partner of choice for countries in the region. In 2021, as part of the Quad, Australia, India, Japan, and the United States established a space working group to share space-based Earth observation data with one another and with other countries in the Indo-Pacific to build a collective understanding of climate change risks and support the sustainable use of oceans and marine resources.¹⁹⁶ In 2023, Japan, South Korea, and the United States held a trilateral dialogue on space security focused on threats in the space domain, the responsible use of space, and national space strategies.¹⁹⁷ Building off of Japan's close bilateral defense partnerships with Australia, the United Kingdom, and the United States (AUKUS), during Prime Minister Kishida's official visit to Washington, D.C., in April 2024, it was announced that AUKUS partners "are considering cooperation with Japan on AUKUS Pillar II advanced capability projects."¹⁹⁸

Japan is also well situated to engage with the Association of Southeast Asian Nations. Several member countries see Japan as the most trusted major power in the region, China as a strategic partner, and the United States as often focused on pressuring countries to pick sides (the United States vs. China).¹⁹⁹ Many of these nations seek to grow their nascent space programs and benefit from the use of space in areas such as Earth observation for maritime domain awareness and natural resource management. They also have a vested interest in preserving the safe, secure, and sustainable use of space. Working with Japan, they can add an important perspective to international discussions on these issues.

Addressing Persistent Challenges

While opportunities abound for cooperation that can deliver material capabilities in support of national policy and security strategies, policymakers will still need to knock down barriers.

Priorities and Time Urgency

It will be important for leaders from Japan and the United States to set clear priorities for their space partnership, harnessing each other's strengths and building interoperable systems to add needed capacity and resiliency for both countries in a timely manner. While both countries have ambitious space architecture plans and have increased resources for space security development and utilization in the past few years, those resources will need to be prioritized and efficiently allocated. Clarity on when capabilities are needed to posture for regional threats will also be important to move their respective institutions and industries into action. Japanese defense strategy highlights the need to take primary responsibility for dealing with invasions by FY 2027.

Designing and building satellites and their ground stations is a multiyear, resource-intensive effort. The timelines for when certain space capabilities are needed to keep pace with the threat will factor heavily into which architectural solutions are pursued and the phasing of those solutions. Fielding a capability in the near term will necessitate a reliance on more mature technologies and existing commercial services where spigots can be turned on. However, investments in longer-term research and development (R&D) efforts are necessary to ensure next-generation capabilities keep apace of the threat.

Government Demand Signal

Regardless of the opportunities for industrial partnerships that present themselves, both U.S. and Japanese industry would benefit from stronger signals and top cover from both governments when attempting to partner to support priority bilateral projects. Much of their reasoning ties to risk and technology export limitations. Several industry representatives conveyed that their companies limit substantial investments in satellite design, R&D, and facilitation unless they have high confidence in a market and customers. Thus, to encourage and support industry collaboration, both governments must create a demand signal and market for such cooperative endeavors, including for interoperability and integration across systems.

There is not currently a government-to-government agreement for industry collaboration on space, although other mission areas have examples that the space relationship may be able to build on. Integrated Air and Missile Defense is one such example. In Integrated Air and Missile Defense, an agreement covers the U.S.-Japan missile defense relationship and provides guidance on cooperation, including information sharing and the codevelopment/coproduction of specific systems, among other items. Frameworks like this help make possible industry collaboration, such as the March 2024 agreement between the Missile Defense Agency and Japan's MOD to codevelop a hypersonic missile defense capability. A similar agreement in space could guide industry prioritization of particular architectures, cooperation models, and information-sharing exchanges. The U.S. DOD and the MOD have a bilateral memorandum of understanding for research, development, test, and evaluation projects, an agreement that commits the nations to increasing opportunities for bilateral collaboration on defense-related emerging technologies. Although space is not mentioned as part of this agreement, this may be an initial avenue to explore space technologies that support defense capabilities such as missile defense.

Finally, there should be follow-through on publicly announced cooperation agreements. During the April 2024 state visit, the United States and Japan announced the negotiation of a Technology Safeguards Agreement (TSA). A TSA would provide a framework for U.S. commercial space launch from Japan and opens further opportunities for space-related technology collaboration within industry.²⁰⁰ Accelerating the completion of negotiations on this TSA will show that both governments have the momentum to follow through with action on high-level talking points that will impact industry.

Data Sharing Policy

On the U.S. side, there is no comprehensive data exchange framework that provides guidance on the technologies, capabilities, and mission knowledge that U.S. industry can share with their Japanese counterparts. Some U.S. government officials express frustration that such a framework may take years to develop, negotiate, and coordinate through administrative processes. Notably, many areas of current U.S.-Japan cooperation take place through idiosyncratic processes and bespoke agreements, not overarching frameworks. Therefore, the speediest way to enable cooperative efforts, such as missile detection and tracking data exchanges, is through a standalone

bespoke agreement negotiated at the senior White House, DOD policy, and State Department levels with their Japanese counterparts.

The areas of cooperation being proposed—such as missile detection and tracking, SDA, and ISRT—involve sensitive technology, capabilities, and mission knowledge that largely fall under U.S. security classification controls, from controlled unclassified information up to higher classification levels. Thus, technical exchanges between U.S. and Japanese companies to explore specific partnership arrangements across satellite development, manufacturing, and architecture interoperability would be limited absent U.S. government clarity on technology transfer and data sharing.²⁰¹ Such direction has been provided for other defense cooperation areas such as SM-3 missiles and Blackhawk helicopters, and both parties have instituted processes and practices to protect sensitive information. A similar approach could be adopted for space security cooperation.

Security

One particularly significant challenge is the difference in cultures of security between the United States and Japan. This includes personnel, information, facility, and cybersecurity, among other areas of difference. The U.S. public and private sectors involved in national security have long-established policies and processes in place and a culture built around protecting sensitive information and advanced capabilities. Security clearances require an intensive look into the backgrounds of personnel employed in the national security ecosystem. However, in Japan security is established more on a case-by-case basis, looking at particular aspects of security for projects and evaluating specific needs.

There is now clear recognition across Japan's government, industry, and academic community that security—cybersecurity, information security, personnel security, and facility security—is paramount to further U.S. space cooperation.²⁰² Implementing security improvements, including recent steps by the Japanese government to institute private sector security clearances and extend classification protections to economic security information, are encouraging and should serve as foundations for getting closer to functional equivalency across several of these security areas.²⁰³

U.S. security experts across government and the private sector could offer training and share best practices. For example, in 2023, Japan's MOD Acquisition, Technology and Logistics Agency developed its first Defense Industrial Security Manual, with advisers from the U.S. Defense Technology Security Administration and the U.S. Defense Counterintelligence and Security Agency.²⁰⁴

There is an acute awareness across the Japanese government and industry of their cyber vulnerabilities. Greater sharing of cyber threats and cyber best practices is the most immediate way to improve Japan's network and cybersecurity. Partnerships with U.S. technology companies, especially in cloud security and cybersecurity, could also be particularly valuable in providing network resiliency and ensuring government continuity of operations. This would also benefit operational space elements, such as the JSOG, to inform their efforts to protect against cyber threats to their networks.

Export Policies

Both countries' export policies and regulations have been barriers to closer space cooperation and industry-to-industry partnerships. U.S. International Traffic in Arms Regulations (ITAR) and the U.S. interpretation of its commitments under the Missile Technology Control Regime (MTCR) hamper bilateral industry cooperation, with both U.S. and Japanese companies that have been encouraged to forge joint space acquisition partnerships citing ITAR and classification as issues. In December 2023, Tokyo revised its principles on the transfer of defense equipment and technology, which is expected to enable the greater export of military technologies and components.²⁰⁵ While a positive development that can further contribute to supply chain diversification and the defense of like-minded partners that are increasingly under threat, the impact of this policy shift for space-related technology exports is unclear. This policy revision demonstrates a significant shift in Japanese defense policy writ large, as well as a political motivation to support such shifts in a range of policy and strategy documents.



U.S. delegates participate in the eighth U.S.-Japan Comprehensive Dialogue on Space.

Photo: Office of Space Commerce

Counterparts and Interagency Communications

Finally, a consistent message from government and industry representatives in both countries is that they often do not know who to talk to on the other side. Observing the U.S. space ecosystem, there are so many points of contact that it is hard to know even what organizations Japanese interlocutors should engage with. As noted by the U.S. Government Accountability Office in a 2016 report, there are “approximately 60 stakeholder organizations” across U.S. national security space.²⁰⁶ Observing the Japanese space ecosystem, responsibility for space security capabilities is spread out across the government. For example, QZSS is with the Cabinet Office, missile detection and

tracking with the MOD internal bureaus, SDA with the JASDF, SATCOM with Joint Staff J6, and ISRT with Japan Defense Intelligence Headquarters. Further, Japan's defense and intelligence institutions remain highly stovepiped. Both sides have found the annual comprehensive space dialogue to be highly valuable in bringing together the countries' space leaders and identifying counterparts, as well as facilitating interagency dialogues within each country. Opportunities to sustain these bilateral and interagency interactions beyond the senior leader dialogue were encouraged.

Summary of Recommendations

Captured here is a summary of the recommendations for space security cooperation described throughout the report.

- **Cultivating human development:** Education and training is foundational to build greater space proficiency across Japan’s defense institutions and supporting industry. Partnerships between the MOD and JASDF and U.S. space institutions across government and academia could include enrollments in U.S. Space 100/200 and Defense Acquisition University courses. Partnerships with new space and technology start-up communities can expand the pipeline of “human resource” talent needed in both Japan’s and the United States’ space ecosystems.
- **Exercising and training the Space Force and joint force:** The United States and Japan should develop a greater understanding of how space capabilities contribute to the joint force, including terrestrial and maritime operations, by integrating a space component into bilateral and multilateral joint exercises and into routine service-specific and joint training. Within the military space community, participation in tabletop exercises and wargames, including multilateral space operations scenarios, as well as greater hands-on operational training, aid in understanding space threats, space operations, and decisionmaking processes.
- **Sharing data and information:** Sharing space and cyber threat assessments is a logical start, especially with the increasing availability of both classified and unclassified threat assessments from the U.S. government, U.S. industry, and academia. Further, SDA data sharing can be enhanced, including the sharing of data from U.S. payloads hosted on Japanese QZSS satellites. SDA information sharing is also an area to expand trilaterally,

building on the U.S.-Japan-South Korea trilateral information exchange model used for data sharing on North Korean missile launches. Updating U.S. policy permissions for data sharing and accelerating implementation of the U.S. DOD's revised space classification guidance can quicken results and build momentum.

- **Emphasizing complementary industrial base strengths:** In formulating partnerships, the United States and Japan should consider the unique and complementary strengths of each nation's industrial base and not limit the partnerships to the space domain. For example, while U.S. industry is strong in the systems integration of military satellite constellations, ground architectures, and scaled-up satellite manufacturing capacity, Japanese industry is a global leader in satellite components, robotics, advanced manufacturing, and shipbuilding and repair. Both nations should look to each other as they seek to increase supply chain resiliency and seek investors to boost their space start-up ecosystems to achieve broader alliance objectives vis-à-vis space.
- **Architecting national security space constellations:** Both countries set out national security space architectures that are ambitious, resource-intensive endeavors, with partnerships being a core tenet of each country's approach. The areas of missile detection and tracking, ISRT, SDA, and SATCOM stand out because of the urgency to have sufficient defensive capabilities deployed by FY 2027 as described in Japan's strategy documents.
- **Missile detection and tracking:** Updating U.S. policy permissions for data sharing and accelerating implementation of revised space classification guidance can quicken results and build momentum. A partnership between the MOD and the U.S. Space Development Agency centered on the tracking and transport capabilities of the Proliferated Warfighter Space Architecture offers an attractive pathway for such phased collaboration and contributes to greater overall capability and resiliency. PWSA ground station deployment in Japan and data sharing, a joint technology and production demonstration program, the exchange of best practices on modular design and tranche-based acquisition and launch, and ground architecture partnerships offer ways to build collective industrial base capacity and overall mission proficiency.
- **Intelligence, surveillance, reconnaissance, and tracking:** The United States and Japan should focus cooperation efforts on adding capacity and resiliency, building greater analytic and multi-intelligence insights, and gearing the overall ISRT architecture toward delivering actionable information on operationally relevant timelines. Another area to explore includes coproduction/license production agreements between U.S. and Japanese space companies and greater access to EO and SAR commercial data and analytic services.
- **Space domain awareness and counterspace:** In contrast to the U.S. challenge of modernizing legacy space traffic management and catalog systems, Japan has an opportunity to take a clean sheet approach to building a common operating picture of the space domain by leapfrogging ahead: leveraging state-of-the-art commercial technology and fusing space surveillance data from both the Japanese and the U.S. governments and commercial sources. The partners should also explore other hosted payload concepts

beyond the USSF SDA-QZSS, consider coordinated operations between their SDA satellites, and explore options to cooperate on space protection. The United States and Japan should further leverage U.S. efforts to reform space program classification to hold in-depth discussions about counterspace concepts of operations.

- **Protected SATCOM:** Japan should be an early international partner in the U.S. SSC project to develop protected tactical waveforms over commercial and international communications satellites. Additionally, discussions on optical interface standards can inform Japan's development of optical communications payloads, and the latter should grow the opportunities for cross linking between communications networks, creating greater resiliency and multipath options for data delivery to users.
- **Investing in digital transformation and cybersecurity:** The United States and Japan should include a digital transformation and cybersecurity component to the bilateral space agenda and encourage the MOD and the JSOG to designate their expanding space architecture, including space operations, C2, and SDA data, as an early candidate for migration to the cloud. This would also facilitate the greater sharing of sensitive mission data and operational information.
- **Leading on policy and in international forums:** Both Japan and the United States, through complementary and coordinated efforts in regional and international forums, can provide strong thought leadership and advance diplomatic initiatives on space sustainability, developing responsible behaviors in space and reducing threats to the peaceful uses of space. Japan should be seen as a regional leader on space security issues and as a partner of choice—along with the United States—for countries in the region.

Persistent challenges that can serve as barriers or limitations to such cooperative efforts will also need to be addressed in key areas:

- **Priorities and time urgency:** It will be important for leaders from both countries to set clear priorities for the space partnership, harnessing each other's strengths and building interoperable systems to add needed capacity and resiliency in a timely manner. Clarity on when capabilities are needed to posture for regional threats will also be important to move their respective institutions and industries into action.
- **Government demand signal:** Both governments must provide clear signaling and direction to industry on specific areas of cooperation and assurance that barriers to cooperation will be addressed, particularly for large-scale bilateral projects on key mission areas like MW/MT and ISRT. Companies will limit investments in substantial satellite design, R&D, and facilitation, unless they have high confidence in a market and customers and that technology-sharing limitations will be addressed.
- **Data sharing policy:** The United States lacks a comprehensive data exchange framework that provides guidance on the technologies, capabilities, and mission knowledge that U.S. industry can share with its Japanese counterparts. The speediest way to enable cooperative efforts, such as missile detection and tracking data exchanges, is through a stand-alone

agreement developed at senior government levels. This authority must operate on a tactically relevant timeline and not be hamstrung by legal reviews concerning the ownership of operational data, binding and nonbinding agreements, and other prosaic details.

- **Security:** Implementing policies and processes to protect sensitive information and advanced technologies, and inculcating them into organizational cultures, is paramount to further U.S. space cooperation. The security imperative encompasses personnel, information, facilities, and cybersecurity, among other areas, and is ripe for U.S. security experts to engage in training and share best practices across government and the private sector.
- **Export policies:** Both countries' export policies and regulations have been barriers to closer space cooperation and industry-to-industry partnerships. U.S. ITAR and interpretation of its MTCR commitments hamper bilateral industry cooperation, with both U.S. and Japanese companies that have been encouraged to forge joint space acquisition partnerships citing ITAR and classification as issues.
- **Counterparts and interagency communications:** A consistent message from government and industry representatives in both countries is that they often do not know who to talk to on the other side. Both sides have found the annual comprehensive space dialogue to be highly valuable in bringing together both countries' space leaders and identifying counterparts, as well as facilitating interagency dialogues within each country. Opportunities to sustain these bilateral and interagency interactions beyond the senior leader dialogue were encouraged.

Conclusion

“For the sake of national security, we will dramatically scale up the use of space systems and ensure the safe and stable utilization of the domain.”

– Japanese prime minister Kishida Fumio, June 2023²⁰⁷

Space is increasingly emphasized in the defense strategies of countries around the world—from global space powers such as the United States, China, and Russia to evolving space leaders such as Japan and France. Aggressive behavior from China and North Korea and Russia’s invasion of Ukraine have caused a shift in the way the Japanese government perceives defensive capabilities and the role space will play.

Japan’s new security emphasis in the space domain is significant, and its recent policies suggest greater Japanese activity in space security in the next five to ten years, spanning from homegrown technology and capability development to expanding international partnerships. Tokyo has an acute awareness of the economic benefits of space and space technology, but developing those capabilities and expanding industrial capacity will likely be a challenge, given the urgency of the need to field operational capabilities in the near to mid term.

As Japan continues to strengthen its space capabilities, its immediate focus continues to be on its space launch sector. Notable milestones will be the success of the H3 and Epsilon S rockets, which aim to cement Japan’s leadership role in space launch, provide a competitive regional alternative to Chinese and Indian launch services, and—in the case of the H3 rocket—play a critical role in the Lunar Gateway program.

In addition to Japan's traditional heavy industry developing launch vehicles, new entrants are exploring robotics, on-orbit servicing, and remote sensing solutions that may further transform Japan's space industry and keep it at the leading edge in certain space technology areas. Sustainability and resource management are two key pillars of Japanese society and policy and are priority areas for space development. Japanese companies are well positioned to develop space solutions to better understand the impacts of climate change and to remove on-orbit space debris.

Tokyo will continue to seek ways to indigenously advance technology and bolster its domestic industrial base. This is to be expected, but it will also be challenging given the urgency of the security environment and the slow pace of development that has defined other areas of Japanese industry. Traditional Japanese space entities, such as JAXA and heavy industry, are expanding their scope from a smaller set of primarily scientific space systems supporting long-term civil missions to larger constellations with a defense and security mission focus. The MOD and SDF must develop new technical and operational expertise in space security to support these new and diversified activities.

Japan will actively expand the role of space capabilities in its security posture going forward. As emphasized in its national defense documents, launch, missile warning, SDA, information gathering, communications, and navigation are all space mission areas with growing defense and national security significance. The establishment of the Space Operations Group also signals a more active role by the SDF in Japan's space enterprise and the continued integration of space capabilities into national security missions. However, while Tokyo is increasing its focus on space and increasing its space spending, its ambitions—especially in the build-out of its space architecture for national security—still surpass its budget. It will be crucial for Japanese policymakers to ensure that Japan's investments into national security space are providing interoperable and secure capabilities that complement the alliance's ability to achieve its objectives in space.

Japan intends to become an increasingly important voice on space in international forums, such as on the threats to the peaceful use of space and responsible behaviors in space, including sustainable use. Tokyo will also continue to be vocal in advocating for stronger and more resilient norms for the use of the space domain to counter adversary threats to the safety of the domain for all users globally.

Additionally, Japanese alignment with the United States on regional threats, the growing role of space in national security, and Japan's increased focus on security opens up greater cooperation opportunities beyond traditional civil space cooperation. However, this security focus will continue to come into tension with Japanese processes and cultural norms, and the two countries will need to continue to break down barriers to cooperation, such as aligning on security and technology protection.

Both nations' governments and industries alike have a multitude of space cooperation opportunities before them that will individually and collectively strengthen their overall national and economic security interests.

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